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Universal properties from quantum many-body dynamics

Joaquín G. Márquez Olguín

Supervisor: Stefano Carignano



**Barcelona
Supercomputing
Center**

Centro Nacional de Supercomputación

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Supervised by: Stefano Carignano

Barcelona Supercomputing Center, Plaça Eusebi Güell, 08034 Barcelona

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The Loschmidt echo—the amplitude for a quantum state to return to itself—provides a dynamical route to obtain universal properties of a critical many-body system. Analytically continuing from real to imaginary time, the echo becomes the partition function of a conformal field theory on a strip. The central charge and operator content of the corresponding conformal-field theory are then encoded in the spectrum of a transfer matrix, obtained by contracting the space–time tensor network along the space direction. The cost of this transverse contraction is governed by the temporal entanglement, which grows only logarithmically at criticality, allowing the method to reach times where conventional evolution algorithms are limited by the entanglement barrier. These results were established for integrable chains, and this thesis extends them to the non-integrable regime. We study a transverse-field Ising model extended by next-nearest-neighbour interactions, which remains critical and in the Ising universality class at weak frustration. We encode its evolution as a matrix product operator, and we resolve the leading spectrum of the resulting non-Hermitian transfer matrix with a block power method that evolves its left and right eigenvectors together. Where the method allows, we recover the conformal signatures: the central charge remains consistent with the Ising value $c = \frac{1}{2}$, and the boundary operator content of the Ising class is reproduced. Throughout, we separate the universal quantities from the non-universal lattice constants that accompany them, and we map where and why the method reaches its limits. *This is a working draft.*

Keywords: Tensor Networks, Transverse Contraction, Many-body Systems, Quantum Dynamics, CFT

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Note on the use of AI

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Acronyms

CFT	Conformal Field Theory	2
DMRG	Density Matrix Renormalization Group	1
ETH	Eigenstate Thermalization Hypothesis	1
FSM	finite-state machine	III
JW	Jordan–Wigner	10
MPO	Matrix Product Operator	6
MPS	Matrix Product State	1
NN	nearest-neighbour	12
NNN	next-nearest-neighbour	2
RDM	Reduced Density Matrix	8
RTM	Reduced Transition Matrix	III
SVD	Singular Value Decomposition	27
TDVP	Time-Dependent Variational Principle	1
TEBD	Time-Evolving Block Decimation	1
TN	Tensor Network	1
tMPO	temporal Matrix Product Operator	6
tMPS	temporal Matrix Product State	14

1 Introduction and Motivation

Following a closed quantum system after being driven far from equilibrium is one of the central questions of modern many-body physics. Simulating such dynamics is difficult, but it can also be revealing: at criticality, the real-time evolution of a simple initial state can expose universal information normally associated with low-energy physics. Our aim with this thesis is to determine how far this dynamical route can be pushed once the system is no longer exactly solvable.

This question is particularly relevant for non-integrable systems, which lack exact analytical solutions. Their dynamics is also intrinsically harder to simulate over long times: after a quench these systems are generically expected to thermalize, acting as their own heat bath so that local observables relax to values fixed only by the energy—the content of the Eigenstate Thermalization Hypothesis (ETH) [1–3]—whereas integrable models, protected by extensively many conserved charges, retain memory of their initial state instead [4]. Analytical results are largely confined to those integrable limits, so almost everything we know about thermalization, scrambling of the quantum information and the onset of chaos in non-integrable models depends on our ability to simulate their dynamics numerically.

Exact numerical simulations are severely limited by the exponential scaling of the Hilbert space, which makes representing a generic many-body wavefunction intractable. To circumvent this bottleneck, the community has turned to Tensor Networks (TNs), a framework that efficiently compresses quantum states by using entanglement as the organizing principle. For one-dimensional systems in equilibrium, Matrix Product States (MPSs) combined with algorithms such as the Density Matrix Renormalization Group (DMRG) have become the state-of-the-art approach [5–7]. Their efficiency comes from the area law of entanglement, which holds for ground states of one-dimensional systems with local gapped Hamiltonians: the entanglement entropy across a bipartition saturates to a constant rather than growing with the subsystem size [8]. Since the required MPS bond dimension is controlled by this entanglement, the ansatz efficiently targets the low-entanglement corner of Hilbert space where these ground states lie.

Simulating dynamics, however, runs into the so-called entanglement barrier. After a quantum quench, the entanglement entropy typically grows linearly in time, approaching volume-law behaviour [9]. Although highly optimized algorithms such as Time-Evolving Block Decimation (TEBD) [10] and the Time-Dependent Variational Principle (TDVP) [11] are very effective at short times, faithfully representing this growth requires a bond dimension that increases exponentially with time, making standard simulations exponentially costly at late times [12, 13].

To overcome this barrier, transverse contraction is a framework that adopts a different perspective [14, 15]. Rather than evolving a state forward in the Schrödinger picture, it represents the full dynamics as a two-dimensional tensor network and contracts it along the original spatial direction. The computational cost is then controlled by the temporal entanglement entropy, defined through a Schmidt decomposition across a cut in time rather than in space. Although its physical interpretation remains an open question, it is computationally well defined. For a generic quench it may grow as strongly as spatial entanglement; nevertheless, the transverse viewpoint can provide a useful way to tackle specific dynamical quantities.

The central quantity we compute throughout is the Loschmidt echo for an infinite, translation-invariant system: the amplitude for a state to return to itself after evolving for a time T [16]. It measures how quickly the evolved state departs from, and how coherently

it returns towards, its initial condition, providing a natural measure of relaxation after a quench. Given its structure, transverse contraction offers an efficient way to compute it by contracting the space–time network directly with a power method [17].

At a conformally invariant critical point, the echo also provides direct access to universal data such as the central charge of the underlying Conformal Field Theory (CFT). In the transverse picture, it is governed by a spatial transfer matrix whose spectrum is predicted by the CFT and which becomes effectively unitary at long times—a phenomenon known as emergent dual unitarity [18]. This behaviour has been demonstrated explicitly for the critical Ising and three-state Potts chains [17], together with the finite-time corrections [19] that we use in Section 7.

The central question is therefore whether these conformal signatures survive the loss of integrability of the model. To address it, we study a self-dual quantum spin chain in the thermodynamic limit that extends the transverse-field Ising model with next-nearest-neighbour (NNN) interactions controlled by a coupling p [20]. For any $p > 0$ it is non-integrable and longer-ranged, making its dynamics costly for conventional methods, yet it remains critical and in the Ising universality class over a wide range of p [20], so the CFT predictions provide a clear target for numerical simulations. We prepare the system in the fully polarized paramagnet state, quench it to the critical point, and use the transverse contraction machinery to compute the Loschmidt echo and the generalized temporal entropies, which we compare with the CFT predictions.

The rest of this thesis is organized as follows. Section 2 defines the Loschmidt echo and develops the CFT framework that relates the transfer-matrix spectrum and generalized temporal entropy to the central charge and operator content. Section 3 develops the transverse-contraction framework: the rotated picture, the temporal states, and the transition matrix from which the generalized temporal entropies are built. Section 4 introduces the ANNNI-type model, establishes its critical Ising universality, and measures the frustration-dependent sound velocity. Section 5 constructs the time-evolution MPO and its non-Hermitian transfer matrix. Section 6 presents the block power method, and Section 7 validates the numerical results with the CFT predictions.

2 Universal Data from the Dynamics

The real-time dynamics of a critical lattice model contains universal information that can be extracted from a simulation. At a continuous critical point, the lattice model forgets its microscopic details and is described by a CFT—a continuum theory so constrained by symmetry that its predictions are fixed by a small set of numbers: the central charge c and the spectrum of scaling dimensions of its operators. These quantities are universal: every model in the same universality class reproduces them, which is exactly what makes them good targets for a numerical method. This section relates the Loschmidt echo to a boundary CFT, states the resulting universal predictions, and identifies the non-universal lattice parameters needed to test them.

2.1 From the Loschmidt echo to a boundary CFT

The object we compute is the Loschmidt echo,

$$\mathcal{L}(T) = \langle \Psi_0 | e^{-iHT} | \Psi_0 \rangle. \quad (1)$$

For an infinite, translation-invariant chain prepared in $|\Psi_0\rangle$, this is the amplitude to return to the initial state after evolving for a time T with the critical Hamiltonian. We take $|\Psi_0\rangle$

to be a product state; its particular choice determines the boundary condition discussed below.

To connect this amplitude with the continuum theory, we rotate the evolution time to the imaginary axis, $T \rightarrow -i\tau$ (a Wick rotation [21]). The real-time propagator e^{-iHT} then becomes the Euclidean weight $e^{-\tau H}$, the same operator that generates the thermal partition function of a two-dimensional statistical system. The matrix element in Eq. (1) can consequently be written as a Euclidean path integral on a slab of width τ , with both edges fixed by $|\Psi_0\rangle$. A state that can terminate a path integral in this way, without introducing any scale of its own, is what a **CFT** calls a conformal boundary state. The echo is therefore the partition function of the critical theory on a long strip whose two edges carry the boundary condition set by $|\Psi_0\rangle$ and whose width is the (imaginary) evolution time. The rotation is used only to obtain the conformal predictions, which are derived on the Euclidean strip; they are continued back to real time before being compared with the simulation. Figure 1 shows this construction, first as a tensor network and then as the strip it represents.

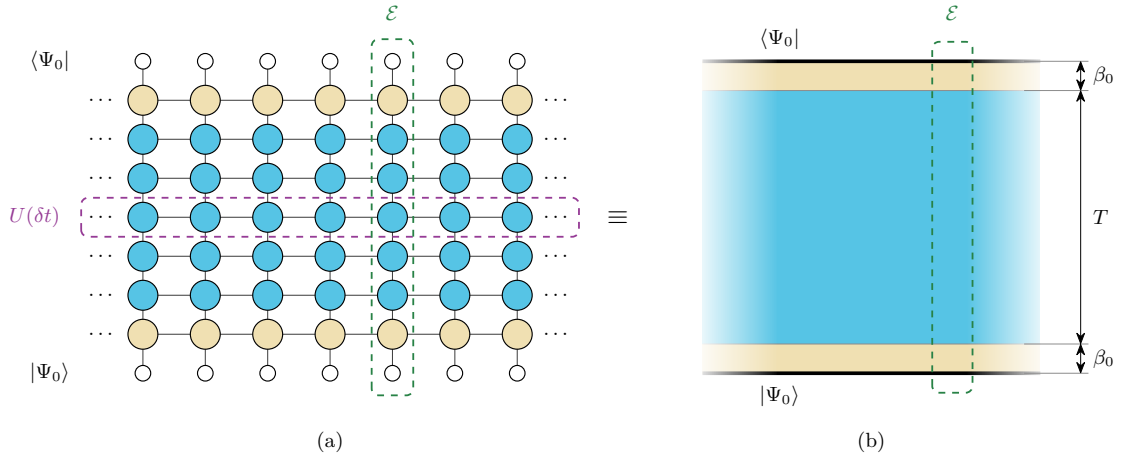


Figure 1: **The Loschmidt echo as a tensor network and as a conformal strip.** (a) Trotterising the evolution expresses the echo as the contraction of a two-dimensional network. Open circles are the product state $|\Psi_0\rangle$ closing the network at either end; each row of filled circles is one Trotter layer $U(\delta t)$, and the cream rows at top and bottom represent the imaginary-time cooling β_0 that brings the bare product state towards a conformal boundary condition. (b) The same object after the Wick rotation, in the continuum. The heavy lines are the edges where $|\Psi_0\rangle$ closes the path integral, the cream bands are the cooling intervals and the blue bulk is the real-time evolution, so the strip spans $T + 2\beta_0$ in the time direction and is infinite in the spatial one.

Two features of this construction are important for the numerics. First, a bare product state carries weight on high-energy configurations and is not itself a conformal boundary state. We therefore cool it in imaginary time for a short interval β_0 , which suppresses this high-energy content and brings the state into the regime where the **CFT** applies. On the lattice, the cooling contributes $N_\beta = 2\beta_0/\delta t$ imaginary-time sites split between the two ends, so the strip spans $T + 2\beta_0$ in the time direction. Second, the initial state is the boundary condition of the strip: each $|\Psi_0\rangle$ realizes a particular conformal boundary condition, whose lightest boundary operator has a dimension x_1 that controls how quickly finite-size corrections decay. For instance, in the Ising class, the disordered (free) boundary has $x_1 = \frac{1}{2}$ and converges quickly, whereas the ordered (fixed) boundary has $x_1 = 2$ and converges slowly. In the σ^x -field convention, the polarized $|X+\rangle$ initial state realizes the free boundary, which is why we quench from it.

We contract the strip in the transverse direction. In the Euclidean description, space and imaginary time are coordinates of the same two-dimensional geometry, so either one can be chosen as the direction of propagation. In the transverse picture, we exchange their usual roles and contract the network along the original spatial direction. This defines a transfer matrix $\mathcal{E}$ that propagates from one spatial column to the next while acting on states supported on a temporal slice. Its tensor-network construction is developed in Section 3.

In two dimensions, conformal symmetry strongly constrains both the spectrum of $\mathcal{E}$ and the entanglement across a cut in the strip [22, 23]. These two observables provide independent routes to the universal data; Appendix A.1 summarises the conformal mappings used to derive them.

2.2 The universal predictions

Confining a CFT to an infinite strip of transverse size β shifts its ground-state energy by an amount set by the central charge. The excited levels above this shift are organised by the operator scaling dimensions x_i . The transfer matrix inherits this structure [17],

$$\mathcal{E} = e^{-a\hat{H}} = \exp\left[-(-\kappa + \pi L_0)g\right], \quad \kappa = \frac{\pi c}{24}, \quad (2)$$

where L_0 is the operator whose eigenvalues list the scaling dimensions, κ is the Casimir number, and $g = a/\beta$ is the ratio of the column spacing to the strip width, with $a = 1$ in lattice units. The width is complex, $\beta = v(2\beta_0 + iT)$, because the regulator contributes imaginary time while the quench contributes real time. Substituting this width and expanding for $\beta_0 \ll T$ shows that the universal contributions enter the eigenvalue phases at order $1/T$ but affect their moduli only at order β_0/T^2 (Appendix A.2). A single diagonalisation can therefore determine both c and the dimensions x_i from the phases.

The second route uses entanglement. In a CFT, the Rényi entropies follow from correlators of twist operators whose scaling dimension is proportional to c [24]. In the transverse setting, the cut runs across the strip in the time direction, and its two sides do not belong to a single quantum state. The entropies are instead built from the left and right temporal MPSs of Section 3; they are therefore generalized entropies [25] and are generally complex. For a cut at time t [17, 19],

$$S_n^{\text{gen}}(t) = s_0 + \frac{i\pi c}{24}\left(1 + \frac{1}{n}\right) + \frac{c}{12}\left(1 + \frac{1}{n}\right)W(t, T), \quad (3)$$

with the chord variable

$$W(x, \ell) = \log\left[\frac{2\ell}{\pi} \sin \frac{\pi x}{\ell}\right], \quad (4)$$

where x is the position of the cut and ℓ is the extent of the system. Evaluating W on the interval $(0, T)$ makes the evolution time play the role of the subsystem size in the spatial formula. The entropy therefore grows as $\log T$, and the bond dimension required for the contraction grows only polynomially with T [12, 13]. We use the Rényi-2 member throughout because it is the least expensive to compute and requires only two copies of the system, making it experimentally accessible [26]. The equilibrium calculations of Section 4 instead use the ground-state von Neumann entropy, which DMRG provides directly from the entanglement spectrum.

Equations (2) and (3) give four predictions, all tested in Section 7. Three concern the transfer eigenvalues μ_i . We work with their logarithms, $\lambda_i = \log(-\mu_i)$, so that $\text{Im } \lambda_i = \arg(-\mu_i)$ is the phase of μ_i measured from the negative real axis. The minus sign shifts

every phase by the same constant π ; this shift cancels in phase differences and is absorbed by a constant when fitting $\text{Im } \lambda_0$ alone.

The generalized-entropy profile. Separating Eq. (3) into real and imaginary parts gives

$$\text{Re } S_n^{\text{gen}}(t) = s_0 + \frac{c}{12} \left(1 + \frac{1}{n}\right) W(t, T), \quad \text{Im } S_n^{\text{gen}} = \frac{\pi c}{24} \left(1 + \frac{1}{n}\right). \quad (5)$$

The real part follows a symmetric chord profile. Plotted against $W(t, T)$, it becomes a straight line whose known slope determines c (Appendix B). The imaginary part is independent of t and gives a second estimate of c without fitting the spatial profile. Finite-time corrections to S_n^{gen} decay as $T^{-1/n}$ [19]; for the Rényi-2 entropy used here this is $T^{-1/2}$, so extracting c requires extrapolation to the long-time limit.

The central charge from the leading eigenvalue. The central-charge and boundary-gap relations are expansions in $1/T$ of the logarithms λ_i , rather than of the eigenvalues themselves. Substituting the strip width into Eq. (2) and expanding gives the following relation [17], derived in Appendix A.2:

$$\frac{\text{Im } \lambda_0}{T} = a_0 + \frac{C}{T^2} + \frac{a_4}{T^4} + \dots, \quad C = -\frac{\kappa}{v} = -\frac{\pi c}{24 v}. \quad (6)$$

Here a_0 and a_4 are non-universal coefficients, while the coefficient of $1/T^2$ carries the Casimir number divided by the velocity. Once v has been measured independently (Section 4.2), the relation $c = 24 v |C|/\pi$ determines the central charge from the eigenvalue phase alone, without relying on the entropy. Because this term is small, the extraction requires a nonzero β_0 and sufficiently long times. In practice we fit $\text{Im } \lambda_0 = AT + B + C/T + D/T^3$ on the unwrapped phase, where B absorbs the branch shift.

The boundary exponent from the first gap. The gap between the leading and first subleading eigenvalues measures the lightest boundary dimension,

$$\text{Im}(\lambda_1 - \lambda_0) = \frac{\pi x_1}{v T}. \quad (7)$$

Once the sound velocity is known, this phase gap determines x_1 . More generally, each subleading eigenvalue corresponds to a boundary dimension x_i through the same relation. We refer to this spectrum—the dominant eigenvalue followed by one subleading eigenvalue for each boundary operator, ordered by increasing phase distance—as the conformal tower, and to its members as tower states.

Emergent dual unitarity. Equation (2) also fixes how the spectrum evolves with T . Unlike the two predictions above, this one is an expansion of the eigenvalues themselves rather than of their logarithms: the moduli of the leading eigenvalues approach a common value while their phases stay distinct,

$$\frac{|\mu_i(T)|}{|\mu_0(T)|} = 1 - \frac{a_i}{T} - \frac{b_i}{T^2} - \dots \xrightarrow{T \rightarrow \infty} 1. \quad (8)$$

Equal moduli with distinct phases indicate that the temporal transfer matrix approaches a unitary operator up to an overall non-universal normalisation. We test this behaviour by following the dominant eigenvalue $\mu_0(T)$ in the complex plane as it moves along a trajectory of nearly constant radius with a winding phase.

These predictions apply to the continuum theory, whereas a simulation contains additional lattice scales and numerical parameters. Section 3 constructs the transfer and transition matrices used to measure the predicted quantities, and Appendix B explains how the universal contributions are separated from the non-universal parameters before the comparison in Section 7.

3 Time Evolution via Transverse Contraction

Section 2 showed that the Loschmidt echo of a critical quench encodes conformal data in the spectrum of a transfer matrix acting along the spatial direction of the corresponding strip. We now construct this transfer matrix and explain how it is used numerically. Transverse contraction was introduced for infinite chains by representing the dynamics with MPS extending along the time direction [14, 15], and was subsequently developed to study temporal entropies [17, 27]. Our calculations use the `ITransverse.jl` implementation developed in [28].

We rotate the space-time tensor network and contract it along the spatial direction. This reformulation maps the contraction to an eigenproblem for the spatial transfer matrix, whose dominant eigenvectors define the temporal boundary states. These states determine the RTM used to compute the generalized temporal entropies. We finally compare two truncation schemes for controlling their bond dimension.

3.1 The rotated network

In the transverse contraction framework, the time evolution of a D -dimensional system is represented by a $(D+1)$ -dimensional tensor network, with the additional dimension corresponding to time. For the one-dimensional chain studied here, the result is a two-dimensional network with no open physical legs. The echo is a scalar amplitude, so this network is analogous to the partition function of a two-dimensional classical model. Its exact contraction is #P-hard in general [29], which requires us to use an approximate representation with a controlled bond dimension (Section 3.4).

The tensor-network representation of the echo follows directly from Eq. (1) and is pictured in Figure 1a. The initial and final states are represented as MPSs, while the evolution operator is decomposed into a sequence of Matrix Product Operator (MPO) layers, one for each Trotter step. Because we require the Hamiltonian to be translation invariant and time independent, the same bulk tensor W appears at every spatial site and time step. Section 5 constructs this tensor for the model with NNN interactions.

In conventional evolution algorithms, successive MPO layers of the time-evolution operator $U(\delta t)$ are applied to the initial MPS $|\Psi_0\rangle$. The resulting spatial entanglement grows linearly with time (volume law) and eventually makes the state intractable at fixed bond dimension [9, 12, 13]. Transverse contraction instead contracts the same network column by column along the spatial direction. The tensors remain unchanged, but their indices exchange roles: the virtual spatial bonds become physical temporal indices, while the physical spatial indices become virtual temporal bonds. Figure 2 illustrates this exchange. The bond dimension of $U(\delta t)$ therefore sets the physical dimension of the temporal problem. Each bulk column defines a temporal Matrix Product Operator (tMPO) $\mathcal{E}$ that acts as a transfer matrix along the original spatial direction, and translation invariance ensures that the same $\mathcal{E}$ is applied at every site.

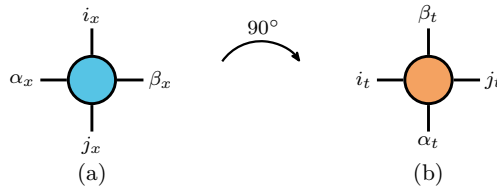


Figure 2: **The rotation exchanges physical and virtual indices.** A rank-4 tensor read (a) along space, with (i_x, j_x) physical and (α_x, β_x) virtual, and (b) along time, where the roles are exchanged.

These objects provide the lattice representation of the strip geometry introduced in Section 2. The strip is infinite along the spatial direction, and applying infinitely many columns on either side of a cut projects the states $\langle L|$ and $|R\rangle$ onto the dominant left and right eigenvectors of $\mathcal{E}$. The resulting fixed points are independent of how the chain is terminated far from the cut. They encode the temporal boundary condition imposed by $|\Psi_0\rangle$ at the top and bottom edges. The strip width is set by the number of temporal sites, including the real-time evolution and the cooling sites associated with the regulator β_0 . The CFT constrains the spectrum of the column transfer matrix $\mathcal{E}$, while the overlap $\langle L|R\rangle$ gives the Loschmidt echo after the spatial contraction.

The row and column transfer operators have different spectral properties. Each row represents the unitary Trotter step $U(\delta t) = e^{-iH\delta t}$ and therefore has eigenvalues of unit modulus. By contrast, the column transfer matrix $\mathcal{E}$ is non-Hermitian and generally has complex eigenvalues with different moduli. The implications of its non-Hermiticity for the left and right eigenvectors are discussed in Section 5 after the operator has been constructed explicitly.

3.2 The dynamics as an eigenvalue problem

After the rotation and Trotter decomposition, the two-dimensional network in Figure 1a is contracted along the spatial direction by repeatedly applying the column transfer matrix between the temporal boundaries. For a chain of N spatial sites,

$$\mathcal{L}(T) = \langle L| \mathcal{E}(T)^N |R\rangle = \sum_i c_i \mu_i(T)^N = c_0 \mu_0(T)^N \left[1 + \sum_{i \geq 1} \frac{c_i}{c_0} \left(\frac{\mu_i}{\mu_0} \right)^N \right], \quad (9)$$

where μ_i are the eigenvalues of $\mathcal{E}$ and the coefficients c_i contain the corresponding overlaps with the boundary states. If μ_0 has the largest modulus, every other contribution is suppressed relative to it by $(|\mu_i|/|\mu_0|)^N$. The thermodynamic limit is therefore determined by the leading eigenvalue, and the intensive Loschmidt rate becomes

$$\ell(T) = - \lim_{N \rightarrow \infty} \frac{1}{N} \log |\mathcal{L}(T)| = - \log |\mu_0(T)|, \quad (10)$$

up to $\mathcal{O}(1/N)$ boundary corrections for a finite open chain. The calculation thus reduces to a spectral problem whose cost is independent of the spatial length.

Equation (9) also identifies the spectral data required for the conformal analysis. The modulus of μ_0 determines the Loschmidt rate, its phase encodes the central charge, and the first spectral gap gives access to the boundary operator content. The generalized temporal entropies instead depend on the dominant left and right eigenvectors. Our calculations must therefore resolve the leading eigenpairs of the non-Hermitian operator $\mathcal{E}$.

To obtain these eigenpairs, we implement the block power method described in Section 6. It evolves a block of k temporal states and resolves the k leading eigenvalues together with their corresponding left and right eigenvectors.

Each iteration increases the bond dimension of the temporal states and must therefore be followed by a truncation. The available truncation schemes are introduced in Section 3.4.

3.3 Temporal states and the RTM

Each application of $\mathcal{E}$ increases the bond dimension of the temporal boundary states. The computational cost is therefore controlled by their temporal entanglement, defined across a cut in the time direction. Section 2.2 showed that in CFT the generalized temporal

entropies grow logarithmically with the evolution time. We now define the object from which these entropies are computed.

The dominant left and right fixed points of the non-Hermitian transfer matrix are distinct, and the echo depends on their overlap $\langle L|R \rangle$. An Reduced Density Matrix (RDM) formed from either state alone therefore does not contain the required information. The relevant object is instead the RTM constructed jointly from the two temporal boundaries,

$$\mathcal{T}_t = \frac{\text{Tr}_{>t} |R\rangle \langle L|}{\langle L|R \rangle}, \quad (11)$$

where $\text{Tr}_{>t}$ traces over the $T - t$ temporal sites above the cut and leaves the t sites below it open. Thus, $\mathcal{T}_t$ acts on the first t temporal sites. The denominator normalizes the matrix so that $\text{Tr} \mathcal{T}_t = 1$. Since $\langle L|$ is not the Hermitian conjugate of $|R\rangle$, $\mathcal{T}_t$ is generally non-Hermitian and may have complex eigenvalues. The consequences of this complex spectrum for truncation are discussed in the next subsection. Figure 3 shows the position of $\mathcal{T}_t$ within the strip geometry.

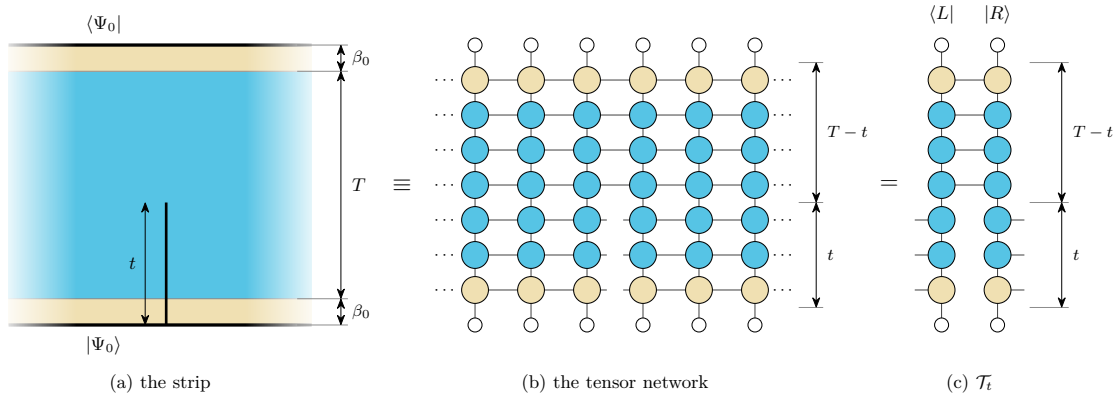


Figure 3: **Position of the reduced transition matrix.** (a) The cut lies on one spatial bond of the strip and extends from the bottom edge to height t . (b) In the tensor-network representation, the horizontal bonds across the cut remain open over the first t temporal sites, while the two sides remain contracted over the remaining $T - t$ sites. (c) Contracting the columns to the left and right of the cut gives $\langle L|$ and $|R\rangle$, respectively. The power method of Section 6 converges these states to the dominant left and right eigenvectors of $\mathcal{E}$. Their contraction over the upper $T - t$ sites gives $\mathcal{T}_t$ in Eq. (11), with the lower t indices left open.

The spectrum of $\mathcal{T}_t$ defines the generalized temporal entropies associated with the transition between the two boundary states. We use the Rényi-2 member of the family introduced in Section 2,

$$S_2^{\text{gen}}(t) = -\log \text{Tr}(\mathcal{T}_t^2). \quad (12)$$

When $\langle L|$ and $|R\rangle$ differ, there is no gauge in which $\mathcal{T}_t$ is locally diagonal, so its spectrum cannot be obtained from a local decomposition. We choose the Rényi-2 entropy because $\text{Tr}(\mathcal{T}_t^2)$ can instead be evaluated directly by contracting two copies of the RTM, without diagonalizing it. This simplification comes with finite-time corrections that must be accounted for when comparing with the conformal predictions, as discussed in Section 2.2.

Once $\langle L|$ and $|R\rangle$ have converged for a target evolution time T , we evaluate Eq. (12) at every internal bond of the temporal chain, corresponding to cuts $t \in (0, T)$. The entropy is generally complex because $\mathcal{T}_t$ is non-Hermitian, so its real and imaginary parts are analysed separately as functions of the normalized cut position.

The temporal region represented by $\mathcal{T}_t$ extends from the lower edge of the strip to the bond at t and therefore has only one endpoint in the interior. Appendix A.1 explains how this edge-anchored geometry fixes the entropy coefficient; if the interval instead had both endpoints in the interior, the logarithmic slope in Eq. (5) would be twice as large. The central charges reported in Section 7 are extracted using the edge-anchored geometry.

3.4 Truncation based on RDM and RTM

We use two complementary truncation schemes based on the **RDM** and the **RTM**. The conventional **RDM** truncation, available in most tensor-network libraries, brings a boundary **MPS** into mixed canonical form and discards the smallest singular values of its orthogonality centre, giving the optimal low-rank approximation to that state [6, 30]. The gauge-invariant scheme implemented in `ITransverse.jl` [28] instead truncates the **RTM** in Eq. (11), which depends on both boundaries. We use the **RTM** scheme for production calculations and retain the **RDM** scheme as a complementary check. Appendix C provides further details on both truncations, including their response to gauge transformations and the conditions for optimal truncation. Here we focus on their main differences: their computational cost and the quantities discarded during truncation.

The **RTM** scheme is substantially cheaper because the two schemes preserve different information. The eigenvalues of the boundary **RDM** are the squared Schmidt coefficients of $|R\rangle$ across the cut. Truncating them retains the directions in which $|R\rangle$ has the greatest weight, irrespective of their overlap with $\langle L|$. The resulting approximation therefore preserves components of $|R\rangle$ that do not contribute to $\langle L|R\rangle$. By contrast, the **RTM** weights each direction by its contribution to this overlap: a large component of $|R\rangle$ carries little weight if it is orthogonal to the corresponding component of $\langle L|$. Thus, the **RTM** can in principle represent the relevant part of both boundaries with fewer retained values and a smaller bond dimension.

The lower cost of the **RTM** scheme comes with a weaker truncation guarantee. For a Hermitian, positive **RDM**, the singular values coincide with the eigenvalues, so retaining the largest Schmidt values gives the optimal approximation at a fixed bond dimension. The **RTM** is instead non-Hermitian, and its eigenvalues may be complex or negative. Although its normalized trace satisfies $\text{Tr } \mathcal{T}_t = \sum_i \lambda_i = 1$, contributions from different eigenvalues may cancel. There is therefore no longer a one-to-one correspondence between the eigenvalues and the singular values. The overlap error can nevertheless be bounded: if the truncation changes the left and right temporal states by at most ϵ_L and ϵ_R in norm, respectively, then the error in their overlap is bounded by $\epsilon_L + \epsilon_R + \epsilon_L \epsilon_R$ [31].

The **RTM** truncation targets contributions to the overlap rather than minimising these norm errors, so the discarded singular values cannot be used directly to evaluate the bound; however, the scheme works in practice for the physical reasons discussed in Appendix C. When its results are noisy or difficult to interpret, we use the more expensive **RDM** scheme as a fallback. Section 7 verifies that the two schemes agree wherever both converge.

4 The ANNNI-type Model

The framework of Sections 2 and 3 requires a quench to a critical point. We study a self-dual quantum spin chain that extends the transverse-field Ising model with **NNN** interactions [20]. For spin-1/2 degrees of freedom, the Hamiltonian is

$$H = - \sum_i \left[\sigma_i^z \sigma_{i+1}^z + \lambda \sigma_i^x + p (\sigma_i^z \sigma_{i+2}^z + \lambda \sigma_i^x \sigma_{i+1}^x) \right], \quad (13)$$

where $\sigma^{x,z}$ are Pauli matrices, λ is the transverse field, and p controls the strength of the additional interactions. The $\sigma^x \sigma^x$ term preserves self-duality under the Kramers–Wannier transformation [20], which fixes the critical point at $\lambda = 1$. We work at this point throughout, so that the conformal predictions of Section 2 apply.

At $p = 0$, Eq. (13) reduces to the integrable transverse-field Ising chain. For $p > 0$, a Jordan–Wigner (JW) transformation no longer maps the Hamiltonian to free fermions, and the model is therefore non-integrable. The origin of the fermionic interaction is detailed in Appendix D.

4.1 Equilibrium criticality and the central charge

Previous finite-size scaling studies find that the chain remains critical and belongs to the Ising universality class up to $p \lesssim 1.5$, with central charge $c \approx 1/2$ [20]. The universality class of this same model has also been established by a different method, which extracts the conformal data directly from the lattice Hamiltonian [32]. We restrict our simulations to $0 \leq p \leq 1.5$, the range over which the model is known to remain critical and in the Ising universality class [20].

We verify the equilibrium universality class over this range using the ground-state entanglement. For a critical open chain of length L , CFT predicts that the von Neumann entropy of a block of length l scales as [24, 30, 33]

$$S_{\text{vN}}(l, L) = \frac{c}{6} W(l, L) + k, \quad (14)$$

where W is the chord variable of Eq. (4) and k is a non-universal constant. The von Neumann entropy is well suited to this calculation because DMRG provides the full ground-state Schmidt spectrum directly, and also has weaker finite-size corrections than the higher-order Rényi entropies [34].

For each value of p , we obtain the ground state of a chain with $N = 400$ sites and evaluate the entropy across every bond. Equation (14) predicts a linear dependence on $W(l, L)$, so the entanglement profile of a single ground state provides an estimate of the central charge. We fit only the middle part of the chain, using the cuts satisfying $0.25L < l < 0.75L$, where boundary corrections are smallest. The resulting profiles are shown in Figure 4.

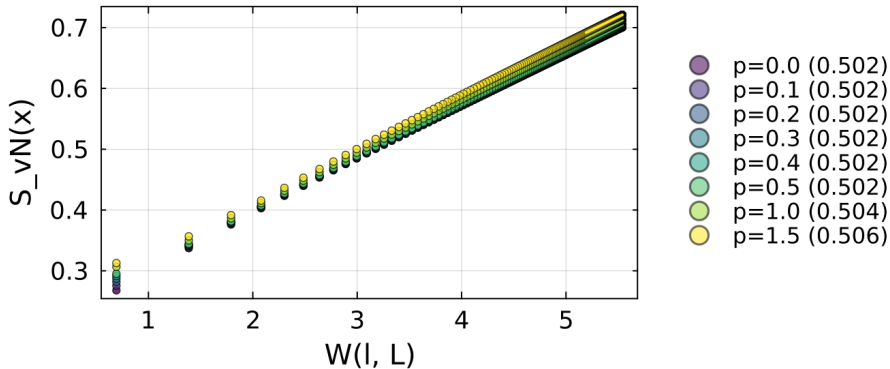


Figure 4: **Effect of NNN interactions to the equilibrium central charge.** Ground-state entanglement profiles at $\lambda = 1$ and $N = 400$, for $p \in [0.0, 1.5]$. The linear dependence on W gives a central charge close to the Ising $c = 1/2$ at every p , with a small common offset from finite-size corrections.

The profiles follow the conformal scaling law at every coupling, and the fitted central charge barely changes with p . Its small common offset from the exact value $c = 1/2$ is consistent with residual finite-size corrections to Eq. (14), since a similar offset is already present at the exactly solvable point $p = 0$. The additional interactions therefore do not change the universality class over the range considered.

The profiles separate at small W , where the cuts lie closest to the open boundaries. Equation (14) is less accurate in this region because the boundaries introduce corrections that depend on the microscopic interactions and therefore on p . This is why these cuts are excluded from the fit. Towards the centre of the chain, these boundary effects weaken and the profiles converge towards the common linear behaviour predicted by CFT. Appendix E.1 gives an independent estimate from the finite-size scaling of the mid-chain entropy and a cross-check using the Rényi-2 entropy.

4.2 Sound velocity

The NNN interaction modifies the propagation of low-energy excitations. At a critical point, the gap closes and the dispersion $\omega(k)$ becomes linear in the lattice momentum k near the gap-closing point. Its slope defines the sound velocity through $\omega(k) = v|k|$, which sets the relativistic scale of the continuum theory. This velocity enters the conformal predictions of Section 2, so it must be determined independently before they can be compared with the dynamical data. For the integrable chain, the exact dispersion $\omega(k) = 2|\sin k|$ gives $v = 2$ [35]; for $p > 0$, we determine v numerically.

We extract $v(p)$ from the equilibrium dispersion of periodic chains, obtained by exact diagonalization, by identifying the lowest excitation with non-zero momentum. Because exact diagonalization is restricted to finite rings, we extrapolate the velocities obtained from five system sizes using their leading $1/N^2$ finite-size dependence. This procedure reproduces the exact result at $p = 0$ and is then applied at each non-zero coupling p .

The velocity grows steadily from 2.000 at $p = 0$ to 10.799 at $p = 1.5$, as shown in Figure 5. We use the independently extrapolated value at each coupling in all subsequent conversions. Although the velocity is non-universal, it is not a fitted parameter in our analysis: it is fixed by the model and determined from the equilibrium spectrum. Appendix E.2 describes the dispersion calculation, extrapolation and validation in detail.

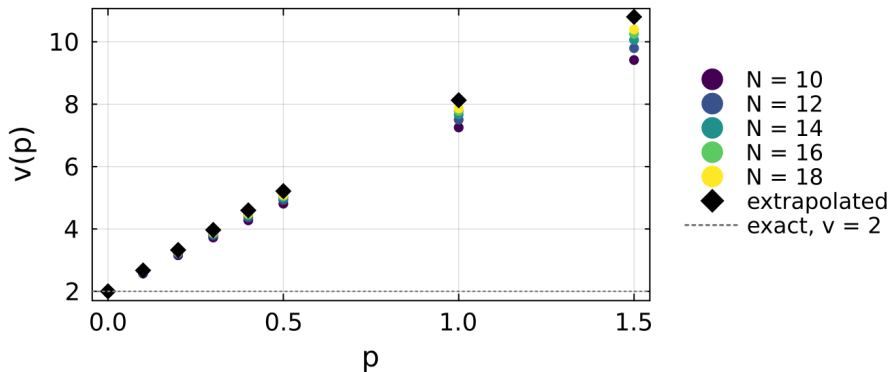


Figure 5: **The sound velocity grows with frustration.** $v(p)$ at $\lambda = 1$, from the equilibrium dispersion of periodic rings of $N = 10$ to 18 sites (coloured circles), together with the values extrapolated to the thermodynamic limit (black diamonds), which are the ones used throughout. The dotted line marks the exact Ising value $v = 2$ at $p = 0$.

5 Building the Time-Evolution MPO

For nearest-neighbour (NN) Hamiltonians, the evolution operator $U(\delta t) = e^{-iH\delta t}$ is usually represented as a Trotter–Suzuki circuit of two-site gates, giving rise to the usual brick-wall structure [13]. However, an interaction between sites i and $i + 2$ cannot be assigned to a single bond within this pattern. To preserve the one-site translation invariance required by the transverse contraction, as discussed in Section 3.1, we instead express the Hamiltonian as a finite-state machine (FSM) and exponentiate the resulting representation directly.

5.1 The Hamiltonian as a FSM

In the FSM representation [36, 37], the Hamiltonian is encoded as an MPO whose local tensor has a block-upper-triangular form [38]. Two identity blocks propagate the initial and final states, while the on-site block D generates local terms. The block row C initiates interactions, the block column B terminates them, and the memory block A propagates unfinished interactions between their endpoints. The virtual index therefore records whether an interaction has not yet begun, is being propagated, or has terminated. For more complicated Hamiltonians, this block structure can be generated automatically from the operator [39]. For the ANNNI-type model, the local tensor has bond dimension $D_W = 5$ and takes the form

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix} = \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\sigma^x & -p\sigma^z & -\lambda\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}. \quad (15)$$

Figure 6 represents Eq. (15) through the allowed paths between the five virtual states. Each node denotes a value of the virtual index, while an arrow from state a to state b , labelled by an operator O , represents the tensor entry $(W_H)_{ab} = O$. The FSM moves from left to right along the chain, beginning in state 1 and ending in state 5. The loop $1 \rightarrow 1$ applies identities until a Hamiltonian term is initiated. A direct transition $1 \rightarrow 5$ applies the on-site field and completes the term at the same site. An NN interaction instead follows either $1 \rightarrow 2 \rightarrow 5$ or $1 \rightarrow 3 \rightarrow 5$: the first transition applies the operator at site i and records whether a σ^z or σ^x endpoint remains to be applied; the second applies the corresponding operator at site $i + 1$ and completes the interaction. The NNN term follows $1 \rightarrow 4 \rightarrow 2 \rightarrow 5$: the first transition applies σ^z at site i , the second inserts an identity at site $i + 1$, and the third applies σ^z at site $i + 2$. After state 5 is reached, the loop $5 \rightarrow 5$ applies identities along the remainder of the chain. Zero entries in W_H correspond to absent arrows, so no other transitions are allowed.

Because the FSM is directional, its local tensor is asymmetric under interchange of the virtual indices. The exponentiated operator inherits this structure, as does the spatial transfer matrix after moving to the transverse picture. The resulting operator $\mathcal{E}$ is generally non-Hermitian and non-normal, $\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$. Its left and right eigenvectors are therefore not related by Hermitian conjugation, and we compute the two temporal boundary states separately.

5.2 Exponentiating the FSM

We seek an approximation $W(\tau) \approx e^{\tau H}$, with $\tau = -i\delta t$, constructed from the blocks of W_H . Neither of the most direct approaches gives a suitable many-body operator: exponentiating

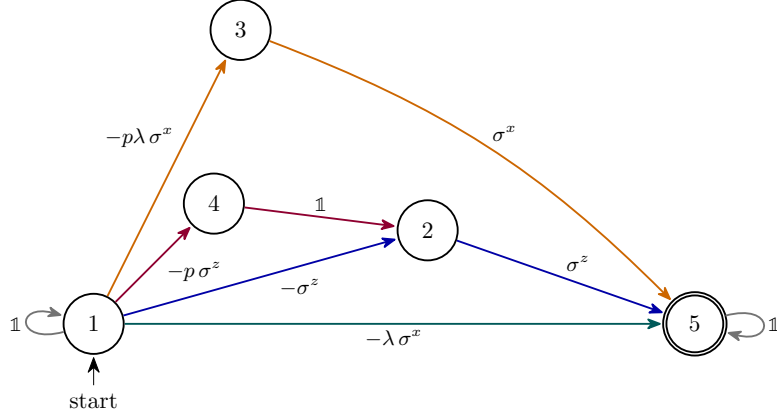


Figure 6: **The Hamiltonian as a finite-state machine (FSM)**. The nodes are the five virtual states of Eq. (15); each arrow is labelled by the operator applied during the transition, and every path from state 1 to state 5 (double circle) generates one term of the Hamiltonian. The arrows are coloured according to the generated term: the transverse field in teal, the **NN** couplings in blue ($\sigma^z\sigma^z$) and orange ($\sigma^x\sigma^x$), and the **NNN** coupling in purple. The grey loops apply identities before and after the Hamiltonian term. State 2 is shared by the blue and purple paths because both terminate with σ^z , keeping the bond dimension at $D_W = 5$.

the local tensor as an ordinary matrix mixes its operator-valued blocks but does not reproduce the exponential of the full Hamiltonian; on the other hand, a finite-order Taylor expansion of $e^{\tau H}$ avoids this local exponentiation, but it is not size-extensive because H^n contains $\mathcal{O}(N^n)$ contributions [40].

We instead use a construction that organises the expansion according to the spatial overlap of the local Hamiltonian terms [40]. Contributions with disjoint support commute and factorise, allowing them to be retained to all orders. Only connected clusters of overlapping terms must be reproduced explicitly. We use the second-order member of this family, denoted *VD2*, whose virtual bond dimension is 13 for the model of Eq. (13). Appendix F gives the full construction, the explicit tensor and the origin of this bond dimension.

Cheaper implementations—with lower bond dimension—include the operators W^I and W^{II} [41]. For a strictly **NN** Hamiltonian, W^{II} is effectively second order. In the present model, however, it omits connected clusters generated by the **NNN** interaction and consequently provides only first-order accuracy. Appendix F.4 demonstrates this difference. We therefore use *VD2* despite its larger bond dimension.

After moving to the transverse picture, the virtual bond dimension of the evolution **MPO** becomes the local physical dimension of the temporal chain. The *VD2* construction therefore has a larger local dimension than W^{II} . Its second-order accuracy permits the larger time step $\delta t = 0.1$ for a given discretisation error, which reduces both the temporal length at fixed physical time and the number of required truncation steps, making the long-time calculations of Section 7 more computationally accessible with *VD2*.

6 The Block Power Method

Section 3 reduced the dynamical calculation to finding the leading eigenpairs of the transfer matrix $\mathcal{E}$. We obtain them with a block power method that evolves several temporal states simultaneously. The method returns the dominant eigenvalues and their left and right eigenvectors, which we then identify and order for comparison with the **CFT** predictions.

Appendix G gives the implementation details.

The right and left eigenvectors of $\mathcal{E}$ satisfy

$$\mathcal{E} |R_i\rangle = \mu_i |R_i\rangle, \quad \langle L_i| \mathcal{E} = \mu_i \langle L_i|. \quad (16)$$

Because $\mathcal{E}$ is non-normal (Section 5.1), its left and right eigenvectors differ, and neither set is orthogonal by itself. Instead, the two sets are biorthogonal, $\langle L_i|R_j\rangle \propto \delta_{ij}$. Here the bra notation does not imply complex conjugation: $\langle L_i|$ is a row vector, and $\langle L_i|R_j\rangle$ denotes the direct contraction of the two temporal networks. We therefore propagate the left states with the transpose $\mathcal{E}^\top$, which implements the second equation in Eq. (16).

6.1 The algorithm

The single-vector power method implemented in `ITransverse.jl` [28] computes only the dominant eigenpair. This is sufficient for computing the Loschmidt rate from the modulus of the dominant eigenvalue μ_0 , the central charge from its phase, and the left and right dominant states ($\langle L_0|, |R_0\rangle$) required for the generalized temporal entropies. It cannot, however, determine the boundary operator content, which is encoded in the phase gaps between μ_0 and the subleading eigenvalues. Accessing these gaps requires the individual subleading eigenvalues and therefore a block power method. Their resolution becomes increasingly demanding with evolution time: from Eq. (7), the i -th phase gap scales as $\pi x_i/(vT)$, decreasing as $1/T$ at fixed coupling and becoming smaller still as the velocity increases with p .

We resolve these closely spaced eigenvalues by iterating a block of k states, which selects the leading subspace collectively. Ordering the eigenvalues by decreasing modulus, after m applications of $\mathcal{E}$ the component along the first discarded eigenvector is suppressed relative to the last retained one by $(|\mu_k|/|\mu_{k-1}|)^m$. The convergence rate is therefore controlled by the separation between the retained subspace and the rest of the spectrum, rather than by gaps between eigenvalues within the block. Once this subspace has converged, a projected eigenproblem separates its individual eigenpairs.

To represent the two subspaces, the block power method evolves k right states $|R_j\rangle$ and k left states $\langle L_j|$, each represented as a temporal Matrix Product State (**tmPS**). We initialise the two blocks independently with random complex tensors. Algorithm 1 summarises one iteration, and Figure 7 shows the tensor-network operations used to apply and project the transfer matrix.

Algorithm 1 Block power method for the k dominant eigenpairs of $\mathcal{E}$.

- 1: $|R_j\rangle, \langle L_j| \leftarrow$ random complex **tmPS**s, $j = 1, \dots, k$
 - 2: **repeat**
 - 3: $|\tilde{R}_j\rangle \leftarrow \text{truncate}_\chi(\mathcal{E} |R_j\rangle), \quad \langle \tilde{L}_j| \leftarrow \text{truncate}_\chi(\langle L_j| \mathcal{E})$ ▷ apply
 - 4: $S_{ij} \leftarrow \langle L_i|\tilde{R}_j\rangle, \quad M_{ij} \leftarrow \langle L_i|\tilde{R}_j\rangle = \langle L_i| \mathcal{E} |R_j\rangle$ ▷ project
 - 5: solve $Mv = \theta Sv$ for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
 - 6: $|R_j\rangle \leftarrow \text{truncate}_\chi(\sum_i V_{ij} |\tilde{R}_i\rangle), \quad \langle L_j| \leftarrow \text{truncate}_\chi(\sum_i U_{ij} \langle \tilde{L}_i|)$ ▷ de-mix
 - 7: normalise every block member
 - 8: **until** the leading Ritz values stop changing within tolerance
-

The first step applies the transfer matrix to each member of the right block and its transpose to each member of the left block,

$$|\tilde{R}_j\rangle = \mathcal{E} |R_j\rangle, \quad \langle \tilde{L}_j| = \langle L_j| \mathcal{E} \iff |\tilde{L}_j\rangle = \mathcal{E}^\top |L_j\rangle. \quad (17)$$

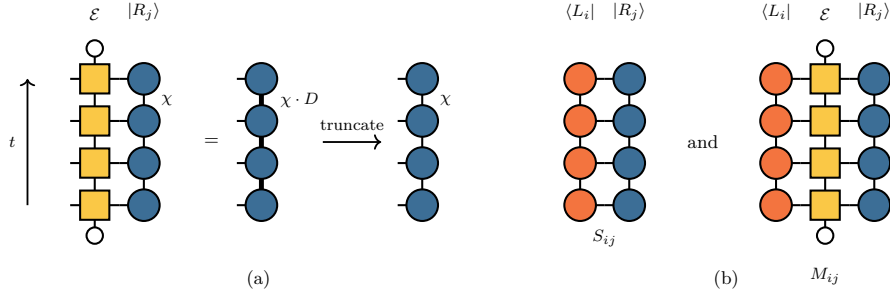


Figure 7: **One iteration of the block power method.** Each tMPS forms a column, while $\mathcal{E}$ is the corresponding column of the space–time network closed at both ends by the product state (small open circles). (a) Applying $\mathcal{E}$ to one member of the right block, as in Eq. (17), increases the temporal bond dimension from χ to χD , where D is the tMPO bond dimension. The result is then truncated back to χ . The left block is treated analogously using the transpose. (b) Contractions defining the two $k \times k$ matrices in Eq. (19). The left chain enters directly as a bra, without complex conjugation.

Applying the tMPO multiplies the bond dimension of each state by D , as shown in Figure 7a. After each application, we truncate the propagated state back to the maximum bond dimension χ using either scheme from Section 3.4 and normalise it, keeping the computational cost of each iteration constant.

The next step projects the eigenproblem onto the current block subspaces. We approximate a right eigenvector by

$$|\psi\rangle = \sum_{j=1}^k v_j |R_j\rangle. \quad (18)$$

The ansatz is a combination of the current block members and not an exact eigenvector, so inserting it into $\mathcal{E} |\psi\rangle = \theta |\psi\rangle$ leaves a residual. We require its contraction with every member of the left block to vanish, so that $\langle L_i | (\mathcal{E} - \theta) |\psi\rangle = 0$. This condition reduces the full eigenproblem to the two $k \times k$ matrices shown in Figure 7b,

$$S_{ij} = \langle L_i | R_j \rangle, \quad M_{ij} = \langle L_i | \mathcal{E} | R_j \rangle, \quad (19)$$

and the generalized eigenvalue problem

$$\sum_j M_{ij} v_j = \theta \sum_j S_{ij} v_j. \quad (20)$$

The overlap matrix S describes the relative orientation of the left and right subspaces, while M is the transfer matrix projected between them. Solving Eq. (20) yields approximations θ_j to $\mu_0, \dots, \mu_{k-1}$, which we order by modulus. It also yields the coefficient matrices V and U used in Algorithm 1; the columns of V specify the combinations of propagated right states associated with the approximate right eigenvectors, and U provides the corresponding combinations for the left block. If S is ill-conditioned, the oblique projection amplifies numerical errors, so we use the pseudoinverse regularisation described in Appendix G when solving Eq. (20).

We then use the Ritz coefficient matrices to separate the propagated blocks into approximate eigenpairs. Before the next iteration, we truncate the resulting states back to χ and normalise them. The iteration stops once the leading Ritz values remain stable within the prescribed tolerance.

The tensor-network operations dominate the computational cost. Most of the cost arises from the sums used to separate the propagated block states. Appendix G.4 quantifies these contributions and describes the measures used to reduce them.

When only the eigenvalues are required, we retain the blocks as orthonormal bases of the leading invariant subspaces instead of assigning individual members to separate eigenvectors. This representation is better conditioned near degeneracies and does not change the Ritz values, as discussed in Appendix G.2. We use it for the late-time spectral measurements in Section 7.

6.2 Selecting and tracking the physical branch

From the k Ritz values returned at each evolution time, we identify the dominant eigenvalue μ_0 and order the remaining tower states. At short times, where the leading spectral gap is wide, μ_0 is unambiguously selected by largest modulus. At later times, emergent dual unitarity drives the tower eigenvalues towards a common modulus (Section 2.2); if the two leading candidates differ in modulus by less than one per cent, we select the one closest in the complex plane to the previously accepted μ_0 , thereby enforcing phase continuity. The subleading eigenvalues are then ordered by increasing phase distance from μ_0 , matching the structure of the conformal tower.

When computing a ladder of evolution times, we initialise the iteration at $T + \Delta T$ with the converged block from T . This continuation tracks the physical branch smoothly from one time step to the next. If the block contains too few states to resolve the required operator content, we expand the block in stages ($k = 4 \rightarrow 6 \rightarrow 8$), initialising each larger subspace from the converged smaller one.

Convergence alone does not guarantee that the iteration reaches the physical fixed point; since the blocks are randomly initialised, different seeds can converge to different, non-physical states at long evolution times. To filter out these artefacts without biasing the results towards CFT predictions, we apply two consistency checks. First, we require $|\mu_0|$ to vary continuously with T , recomputing any discontinuous point with a fresh random seed. Second, when extracting eigenvectors for the entropy, we repeat the calculation until two independent runs agree. Since two random seeds can occasionally converge to the same unphysical fixed point, reproducibility alone is insufficient; we therefore also require the accepted entropy to vary smoothly with neighbouring evolution times.

We determine the upper limit of each time window from the winding of the leading eigenvalue. Along the physical branch, the phase of μ_0 advances at an approximately constant rate with T , so any persistent departure from this rate signals that the branch is no longer resolved, and therefore sets the boundary of the usable fitting window. Appendices G.6 and G.7 give the corresponding selection safeguards and conditioning diagnostics.

7 Results

We now test the CFT predictions of Section 2.2 against simulations of the critical ANNNI-type chain (Eq. (13) with $\lambda = 1$), quenched from the polarized state $|X+\rangle$. We vary the NNN coupling from the integrable point, $p = 0$, up to $p = 0.5$. Because extracting the universal data reliably requires a sufficiently wide time window, and the simulation cost grows significantly with p , we restrict the dynamical calculations to $p \leq 0.5$ for simplicity and because it suffices for extracting the universal properties from the critical system.

The analysis divides into two complementary routes. The entropy route uses the transfer-matrix eigenvectors to compute the generalized Rényi-2 entropy (Eq. (12)) and compares it with the conformal profile (Eq. (5)). The spectral route uses the eigenvalues alone to verify emergent dual unitarity, extract the central charge from the phase curvature

of the leading eigenvalue (Eq. (6)), and determine the boundary exponent from the first spectral gap (Eq. (7)). In both routes, we evaluate the integrable Ising point ($p = 0$) alongside the non-integrable cases ($p > 0$) to test whether the conformal predictions remain valid in the presence of the **NNN** interaction. Table 1 collects the universal data obtained from both routes, and the rest of this section explains how each value is measured.

Unless stated otherwise, all simulations use the second-order **VD2 MPO** of Section 5 with time step $\delta t = 0.1$, cooling regulator $\beta_0 = 0.2$ ($N_\beta = 4$), and maximum temporal bond dimension $\chi = 64$. We convert all spectral quantities into universal data using the independently determined sound velocities $v_\infty(p)$ from Section 4.2, with fitting conventions and controls detailed in Appendix E.4.

p	v_∞	window	spectral c	x_1	entropy-route c
0	2.000	$T \leq 20$	0.45	0.498	[OPEN]
0.1	2.670	$T \leq 9$	0.47	0.497	[OPEN]
0.3	3.967	$T \leq 6$	0.49	0.493	[OPEN]
0.5	5.212	$T \leq 5$	0.53	0.488	[OPEN]

Table 1: **Universal data against the **NNN** coupling.** The spectral central charge comes from Eq. (6) on ladders sampled at $\Delta T \leq 0.5$ inside the window, with the branch constant pinned. The boundary exponent is the first phase gap converted through Eq. (7). Appendix E.4 gives the fit windows and the controls behind these numbers. The entropy-route central charge is the extrapolated plateau of Section 7.1.

7.1 The entropy route

At the integrable point, the generalized temporal entropy follows the conformal prediction established for the Loschmidt echo of the critical Ising chain [17]. We find that this behaviour persists when the **NNN** interaction is introduced. Figure 8 shows the Rényi-2 profiles at $p = 0.1$. The real part follows the chord dependence in Eq. (5), while the imaginary part is approximately constant in the interior of the strip. For this comparison, we fix $c = \frac{1}{2}$ and adjust only the non-universal constant s_0 , following the analysis of the integrable chain. Appendix E.3 presents the corresponding profiles at the remaining couplings.

At every simulated time, the measured plateau lies above the asymptotic value $\pi c/16$. This offset is the finite-time correction discussed in Section 2.2 [19]. Because it decays slowly with T , the asymptotic plateau must be obtained by extrapolating a sequence of evolution times. A short time series cannot distinguish the decay from a constant offset and therefore does not constrain the asymptote. Including the finite-time correction gives $c = [\text{OPEN}]$ at the integrable point, consistent with the exact value $\frac{1}{2}$, and $c = [\text{OPEN}]$ at $p = 0.1$. In contrast to Figure 8, where c is fixed to test the predicted profile, these extrapolations determine c from the data. The results are included in Table 1.

We use the imaginary plateau rather than the real-part chord slope to estimate the central charge. At each evolution time, Eq. (5) predicts the plateau value directly. By contrast, extracting c from the real part requires a two-parameter fit of the chord coefficient and the non-universal offset s_0 , making the result more sensitive to the interval of the strip included in the fit.

The accessible time window decreases as p increases, leaving fewer points with which to determine the finite-time decay. The latest accessible times also require the repetition procedure described in Section 6.2. When the available window is too short to constrain

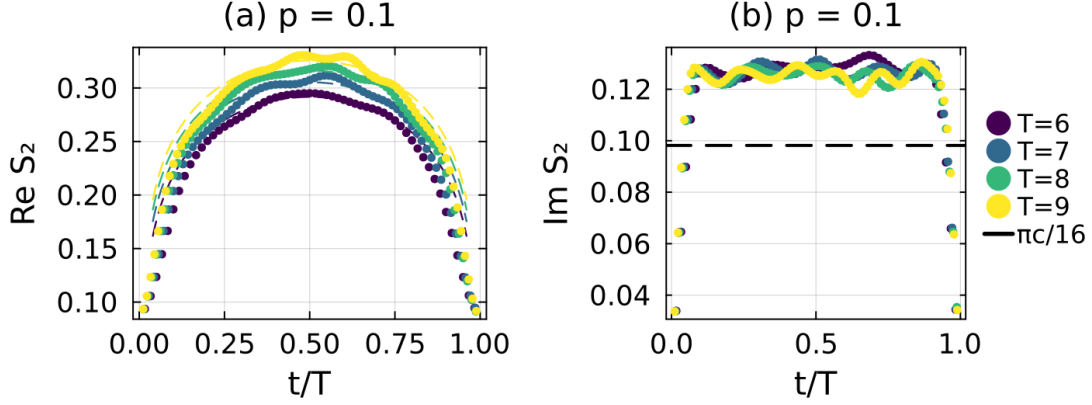


Figure 8: **The generalized temporal entropy at $p = 0.1$.** Each colour denotes one evolution time. (a) Real part as a function of t/T . The dashed curves show the conformal profile in Eq. (5) with $c = \frac{1}{2}$. (b) Imaginary part, which is approximately constant in the interior of the strip. Its plateau lies above the asymptotic value $\pi c/16$ (dashed line) because of finite-time corrections that decrease with increasing T .

the extrapolation, we do not report an entropy-based central charge and instead use the spectral method below.

7.2 The spectral route

Using the eigenvalue-only method of Section 6.1, we compute the spectrum up to $T = 24$ at the integrable point and up to $T = 20$ at the remaining couplings. Only the shorter fitting windows listed in Table 1 are used to extract universal quantities. Figure 9 shows the trajectory of μ_0 in the complex plane after normalising each time series by its mean modulus. The phase winds throughout the simulated interval, while the modulus varies by less than one per cent. This behaviour is consistent with the emergent dual unitarity described in Section 2.2. The corresponding central-charge fits are shown in Figure 10.

We extract the central charge from the phase curvature using Eq. (6). Each time series is sampled with $\Delta T = 0.5$ because the fit is constrained by both the extent of the window and the number of points within it. The fitting conventions are given in Appendix E.4.

Table 1 summarises the results. The spectral estimates remain within approximately ten per cent of the Ising value at every coupling and are consistent with $c = \frac{1}{2}$ within the resolution of the fits. The equilibrium chord analysis of Section 4.1 gives $c \approx 0.502$ over the same range. These determinations use independent data: the equilibrium estimate is obtained from the ground state, whereas the spectral estimate follows from the real-time dynamics.

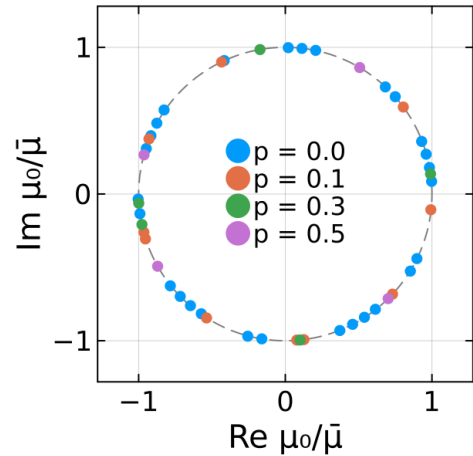


Figure 9: **Trajectory of the normalised leading eigenvalue.** At each coupling, μ_0 is divided by its mean modulus over the time series. The phase winds with T , while the weak variation of the modulus keeps the points close to the dashed unit circle.

The sound velocity affects the calculation in two competing ways. Finite-time corrections are controlled by vT , so their relative magnitude decreases as v grows at fixed T . However, the phase gaps also scale as $1/(vT)$ and therefore become more difficult to resolve. In addition, converting the fitted curvature into a central charge multiplies its numerical uncertainty by v . These resolution requirements cause the fitting windows in Table 1 to shorten with increasing p , even though finite-time corrections within those windows become smaller.

The Ising boundary tower persists when the NNN interaction is introduced. As shown

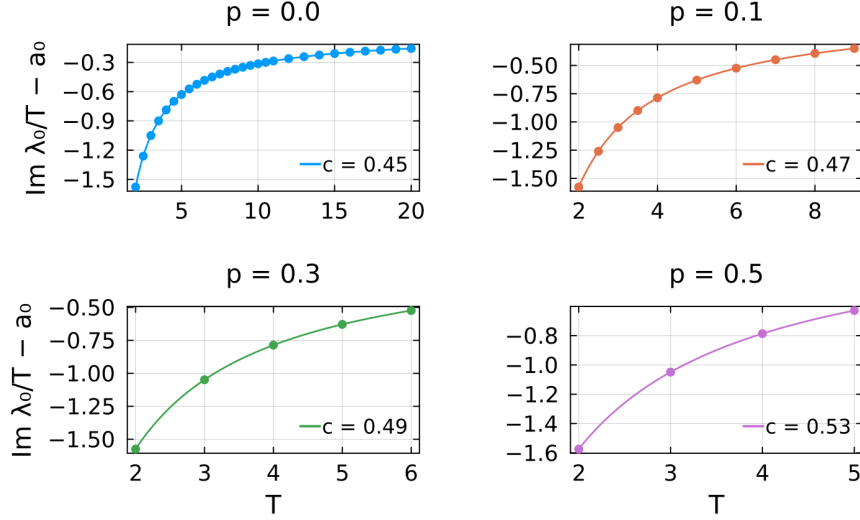


Figure 10: **Central charge from the phase of the leading eigenvalue.** Fits of Eq. (6) to the phase of μ_0 , shown separately for each coupling. The fitted constant is subtracted to display the time-dependent contribution. The curvature determines the central charge reported in each legend.

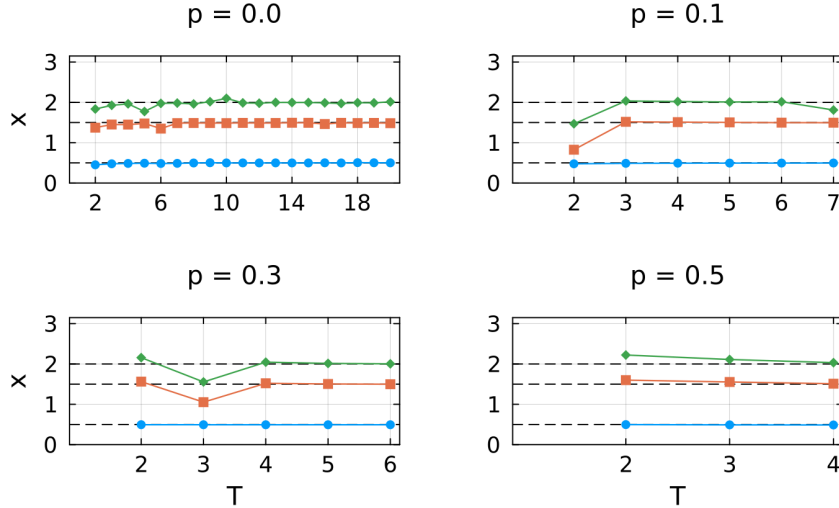


Figure 11: **Boundary dimensions as a function of evolution time.** The three leading dimensions at each coupling are obtained from the phase gaps through Eq. (7). The series show x_1 (circles), x_2 (squares), and x_3 (diamonds), in ascending order; dashed lines mark the Ising values $\frac{1}{2}$, $\frac{3}{2}$, and 2 . The leading dimension remains close to $\frac{1}{2}$, while the higher dimensions approach their predicted values as T increases. The $p = 0$ calculation uses $k = 4$, and the remaining couplings use $k = 8$.

in Figure 11 and Table 1, the leading dimension remains close to $\frac{1}{2}$ at every coupling. At fixed evolution time, its deviation from this value decreases as p increases, consistent with finite-time corrections controlled by vT .

Resolving the higher levels requires a larger block. With $k = 8$, the first three dimensions approach the Ising values $\frac{1}{2}$, $\frac{3}{2}$, and 2 at every non-integrable coupling shown in Figure 11. The remaining Ritz values do not converge sufficiently within the accessible time windows, so we do not assign them to higher members of the conformal tower.

8 Conclusions and Outlook

The question that motivated this thesis was whether the conformal signatures accessible through transverse contraction persist when integrability is broken. For the self-dual ANNNI-type chain and the dynamical couplings studied here, the results remain consistent with the Ising universality class. The leading transfer-matrix eigenvalue has an approximately time-independent modulus and a winding phase, as predicted by emergent dual unitarity. Its phase curvature gives a central charge compatible with $c = \frac{1}{2}$ within the resolution of the fits, while the leading phase gaps reproduce the Ising boundary dimensions (Table 1). These results indicate that the conformal structure used by the transverse method is not restricted to the integrable point. Within the critical regime considered, the NNN interaction modifies the non-universal scales without changing the resolved universal content.

Obtaining these results required a transfer-matrix construction suitable for longer-range interactions. We represent the Hamiltonian as a FSM, exponentiate it with a second-order scheme, and rotate the resulting MPO into a column of the space-time network. The standard second-order construction for NN Hamiltonians becomes first order in the presence of NNN terms, whereas the VD2 construction retains second-order accuracy by including the corresponding three-site clusters. We then determine the leading spectrum with a block power method that propagates left and right temporal states together and preserves their biorthogonal pairing. An eigenvalue-only variant replaces the individual eigenvectors by well-conditioned bases of the leading subspaces when the calculation does not require the eigenvectors themselves.

The equilibrium analysis provides the non-universal input and an independent reference for the dynamical results. DMRG calculations give a central charge consistent with the Ising value up to $p = 1.5$, while exact diagonalisation of finite rings determines the sound velocity at each coupling (Section 4). This velocity converts the transfer-matrix phase gaps into scaling dimensions and must therefore be determined independently. The equilibrium and dynamical estimates of the central charge are consistent within the numerical resolution of the dynamical fits, although the latter deviate from $\frac{1}{2}$ by up to approximately ten per cent over the reported range.

The entropy and spectral routes have different numerical ranges because they require different parts of the eigensystem. First-order perturbation theory shows that eigenvector mixing is inversely proportional to the separation between neighbouring eigenvalues, whereas the corresponding eigenvalue shift contains no spectral-gap denominator. As the conformal phase gaps decrease with increasing T , separating individual eigenvectors within the converged block becomes ill conditioned. The entropy calculation requires the dominant left and right eigenvectors and consequently loses reliability at earlier times. The spectral calculation requires only the Ritz values of the leading subspace and remains usable over a wider range.

This conditioning argument explains the greater stability of the eigenvalues, but it

does not fully determine when an entropy calculation fails. The phase rigidity decreases with evolution time and more rapidly at stronger coupling, yet its value at the last reliable time varies by four orders of magnitude across the couplings studied. Failures are also intermittent and depend on the random initial states: a calculation can fail at one time, converge at the next, or be recovered by repeating it with a different seed. Moreover, the bond dimension, truncation cutoff, and eigenvalue-selection rule were chosen through internal consistency tests rather than a systematic optimisation. The reported time windows should therefore be understood as the reach of the present numerical configuration, not as fundamental limits of transverse contraction.

The sound velocity provides a useful indicator of this numerical reach. Boundary dimensions are extracted from phase gaps proportional to $1/(vT)$, so increasing v compresses the spectrum and raises the resolution required at a fixed evolution time. Since v can be measured independently in equilibrium, this compression can be estimated before performing the dynamical calculation.

Several improvements follow directly from these limitations. A systematic study of the bond dimension, truncation cutoff, and eigenvalue selector is needed to determine which parameter controls convergence in each regime. The computational cost is dominated by the de-mixing and compression of the propagated block states, so reducing this cost would extend the accessible evolution times. With these improvements, the dynamical central charge $c(p)$ and leading boundary dimension $x_1(p)$ could be followed towards the boundary of the critical phase, where the equilibrium calculations remain consistent with Ising criticality but the present dynamical windows are too short for reliable fits.

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A Conformal-field-theory derivations

This appendix summarises the standard **CFT** results used in Section 2: the chord form of the two-point function, the strip quantisation underlying the transfer-matrix spectrum, and the entropy coefficients. Full derivations are given in Refs. [23, 24]; here we focus on their application to the present calculation.

A.1 Conformal maps, the two-point function and the entropy coefficients

In two dimensions, the Euclidean coordinates are written as $z = x + i\tau$ and $\bar{z} = x - i\tau$. A primary operator ϕ has conformal weights $(h, \bar{h})$, which determine its transformation under the two chiral sectors. Their sum,

$$x = h + \bar{h} \quad (21)$$

is the scaling dimension used in the main text. On the plane, the two-point function has the power-law form

$$\langle \phi(z_1) \phi(z_2) \rangle = \frac{1}{(z_1 - z_2)^{2h} (\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}. \quad (22)$$

The transformation law of a primary field then determines the correlator on conformally related geometries. Under the logarithmic map from the plane to a cylinder of circumference β , Eq. (22) becomes [23, 24]

$$\langle \phi(\omega_1) \phi(\omega_2) \rangle \propto \left[\sinh \frac{\pi(\omega_1 - \omega_2)}{\beta} \right]^{-2h} \left[\sinh \frac{\pi(\bar{\omega}_1 - \bar{\omega}_2)}{\beta} \right]^{-2\bar{h}}. \quad (23)$$

Analytic continuation to real time, $\beta \rightarrow iT$, replaces the hyperbolic sine by a sine and produces the strip geometry relevant to the quench. The resulting time-dependent correlator gives the chord variable in Eq. (4).

The entropy coefficients follow from the same two-point function. The Rényi entropy of an interval $\mathcal{A}$ is obtained from the moments $\text{Tr } \rho_{\mathcal{A}}^n$, with the von Neumann entropy recovered by analytic continuation to $n \rightarrow 1$. In the replica construction, each endpoint of $\mathcal{A}$ in the interior of the system introduces a twist operator with scaling dimension [24]

$$\Delta_n = \frac{c}{12} \left(n - \frac{1}{n} \right). \quad (24)$$

For a bulk interval, both endpoints lie in the interior. The moments therefore follow the two-point scaling $\text{Tr } \rho_{\mathcal{A}}^n \sim \ell^{-4\Delta_n}$, which gives

$$S_n = \frac{c}{6} \left(1 + \frac{1}{n} \right) \log \ell + \text{const}, \quad S_{n \rightarrow 1} = \frac{c}{3} \log \ell + \text{const}. \quad (25)$$

The temporal bipartitions used in this thesis instead terminate at an edge of the strip and have only one endpoint in the interior. The replica construction then contains a single twist operator, reducing the logarithmic coefficient by a factor of two. The coefficient is therefore $\frac{c}{12} (1 + \frac{1}{n})$, which becomes $c/6$ as $n \rightarrow 1$ and $c/8$ at $n = 2$, rather than the bulk value $c/3$. Using the bulk coefficient would consequently overestimate the fitted central charge by a factor of two.

A.2 The transfer-matrix spectrum from the stress tensor

The stress-energy tensor determines the response of the theory to coordinate transformations. In two dimensions, its modes L_n generate conformal transformations and satisfy the Virasoro algebra, whose central term is proportional to c [23]. The generator L_0 controls scale transformations, and its eigenvalues on primary operators are their conformal weights. Its spectrum therefore encodes the operator content of the theory.

Quantisation on a strip of transverse width β makes the conformal anomaly explicit. The generator of translations along the strip is obtained by integrating the stress tensor. Under the conformal map from the plane, it acquires a geometric contribution proportional to c [17],

$$\hat{H} = \frac{\pi}{\beta} \left(L_0 - \frac{c}{24} \right). \quad (26)$$

Equation (26) contains both forms of universal information. The constant $-c/24$ is the Casimir contribution and determines the central-charge term, while the spectrum of L_0 gives the scaling dimensions. Since the transfer matrix advances the network by one spatial column, $\mathcal{E} = e^{-a\hat{H}}$, where a is the lattice spacing. Substituting Eq. (26) gives the spectrum in Eq. (2), with $g = a/\beta$ and $a = 1$ in lattice units. The dominant eigenvalue contains the Casimir term, whereas ratios and phase differences involving the subleading eigenvalues encode the operator dimensions.

The effective strip width contains both imaginary- and real-time contributions. The regulator supplies imaginary-time intervals at the two boundaries, the quench supplies the intervening real-time interval, and the velocity converts these durations into lengths:

$$\beta = v(2\beta_0 + iT), \quad \frac{1}{\beta} \simeq \frac{1}{v} \left(\frac{2\beta_0}{T^2} - \frac{i}{T} \right), \quad (27)$$

where the second expression is expanded to first order in β_0/T . Substitution into Eq. (2) gives

$$\log \mu_i \simeq \frac{i\pi}{vT} \left(x_i - \frac{c}{24} \right) - \frac{2\pi\beta_0}{vT^2} \left(x_i - \frac{c}{24} \right). \quad (28)$$

The phases contain the universal data at order $1/T$, whereas the moduli depend on the regulator at order β_0/T^2 . Setting $x_i = 0$ for the identity gives Eq. (6), and subtracting two eigenvalues gives Eq. (7). For small β_0 , the moduli therefore vary weakly with time, producing the emergent dual unitarity tested in Section 7.

B Universal versus non-universal constants

The predictions of Section 2 apply in the continuum limit, whereas the simulations use a microscopic lattice at finite evolution time T , bond dimension χ , Trotter step δt , and regulator β_0 . The measured quantities therefore contain universal and non-universal contributions. The central charge c and scaling dimensions x_i are fixed by the universality class and determine the coefficients in Eqs. (5)–(8). By contrast, the regulator and associated entropy offset s_0 , the Trotter step, the sound velocity v , and the overall transfer-matrix normalisation depend on the lattice construction or numerical method. Separating these contributions determines which conformal predictions can be tested directly.

The sound velocity is determined independently. We use the extrapolated value $v_\infty(p)$ from Section 4.2 in all subsequent conformal expressions. Since the eigenvalue phases depend on $\pi x/(vT)$, this velocity converts each measured phase gap into a scaling dimension x_i .

Universal and non-universal contributions affect the spectrum differently. The overall normalisation of the transfer matrix is fixed by the lattice construction rather than the continuum theory. Each network column contains one Trotter layer and the cooling rows, so this scale depends on δt , β_0 , and the normalisation convention for the local **MPO** tensor. Its explicit form is unnecessary because it multiplies all eigenvalues equally, rescaling their moduli without changing their phases. We therefore extract scaling dimensions from phase differences and use the moduli only through ratios or asymptotic trends.

A ratio of phase gaps also removes the velocity and regulator dependence:

$$\frac{\text{Im}(\lambda_2 - \lambda_0)}{\text{Im}(\lambda_1 - \lambda_0)} = \frac{x_2 - x_0}{x_1 - x_0}, \quad (29)$$

The normalisation, β_0 , and v cancel between numerator and denominator. This ratio therefore probes the relative operator content without external calibration.

The entropy slope in Eq. (5) cannot be reduced to such a ratio. At finite $(T, \chi, \delta t)$, it differs from its continuum value:

$$\text{slope}(p) = \frac{c(p)}{8} K(T, \chi, \delta t, \beta_0) + R(p; T), \quad (30)$$

Here K collects finite-time, truncation, and Trotter corrections shared by two models in the same universality class when evaluated with matched parameters. The term R contains model-dependent subleading corrections whose amplitudes depend on the operator content and boundary condition [19]. At the integrable point, where $c = \frac{1}{2}$, the measured $\text{slope}(0)$ determines the shared factor K . Taking the ratio then cancels K , but not R :

$$c(p) \approx \frac{1}{2} \frac{\text{slope}(p)}{\text{slope}(0)}. \quad (31)$$

This calibration assumes that the two models share a universality class, which is itself under investigation. The residual term R sets the uncertainty, so the calibrated value provides a consistency check rather than an independent determination. Alternatively, the **CFT** fixes the functional form of the leading corrections [19], allowing them to be included explicitly and c to be extracted without a reference model. This absolute method requires a sufficiently long time window to separate the correction from the leading term, whereas calibration can be applied at shorter times but requires an integrable reference.

C Truncation schemes

The general **TN** framework, including diagrammatic notation, matrix product representations, canonical forms, and standard algorithms, is reviewed in Refs. [6, 7]. Here we present the linear algebra underlying the truncation choice in Section 3.4. We explain why Schmidt-value truncation is optimal for a single state, how gauge freedom distinguishes physical from representation-dependent spectra, and how the construction extends to the **RTM**.

C.1 Gauge freedom and canonical forms

An **MPS** representation is not unique. On any virtual bond, one may insert $\mathbb{1} = XX^{-1}$ and absorb the two factors into neighbouring tensors, $A^{[n]} \rightarrow A^{[n]}X$ and $A^{[n+1]} \rightarrow X^{-1}A^{[n+1]}$, without changing the represented state. A canonical form fixes this freedom. Successive QR or Singular Value Decomposition (**SVD**) decompositions make the tensors isometric

from the left, $A^\dagger A = \mathbb{1}$, or from the right, $BB^\dagger = \mathbb{1}$, leaving a single centre tensor that carries the non-trivial weight. The full construction is given in Refs. [6, 7].

A physical quantity must be invariant under this gauge transformation. Quantities that depend on individual tensors rather than on the represented state are gauge dependent and cannot define an invariant truncation criterion. For a single state, the canonical form fixes the gauge and exposes the physical Schmidt values. In the transverse construction, the distinction becomes essential because the left and right states are different.

C.2 Schmidt values and optimal truncation

Consider a cut dividing the chain into a block $\mathcal{A}$ and its complement $\mathcal{B}$. Grouping the physical indices on each side into one composite index turns the wavefunction into a matrix, whose **SVD** gives the Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^{\chi} s_{\alpha} |\alpha\rangle_{\mathcal{A}} \otimes |\alpha\rangle_{\mathcal{B}}, \quad s_{\alpha} \geq 0, \quad \sum_{\alpha} s_{\alpha}^2 = 1, \quad (32)$$

where the states on each side form orthonormal bases. In mixed canonical form, this decomposition is represented on the cut bond by a diagonal matrix of Schmidt coefficients. The corresponding **RDM** is diagonal in the same basis, with eigenvalues $p_{\alpha} = s_{\alpha}^2$ that determine the entanglement across the cut.

Reducing a bond dimension from χ to $\chi' < \chi$ retains the χ' largest Schmidt coefficients. By the Eckart–Young–Mirsky theorem, this gives the closest state to $|\psi\rangle$ among all states with Schmidt rank at most χ' . The squared norm error equals the discarded weight,

$$\varepsilon = \|\psi\rangle - |\tilde{\psi}\rangle\|_2^2 = \sum_{\alpha > \chi'} s_{\alpha}^2. \quad (33)$$

Figure 12 shows why the truncation can be performed locally in canonical form.

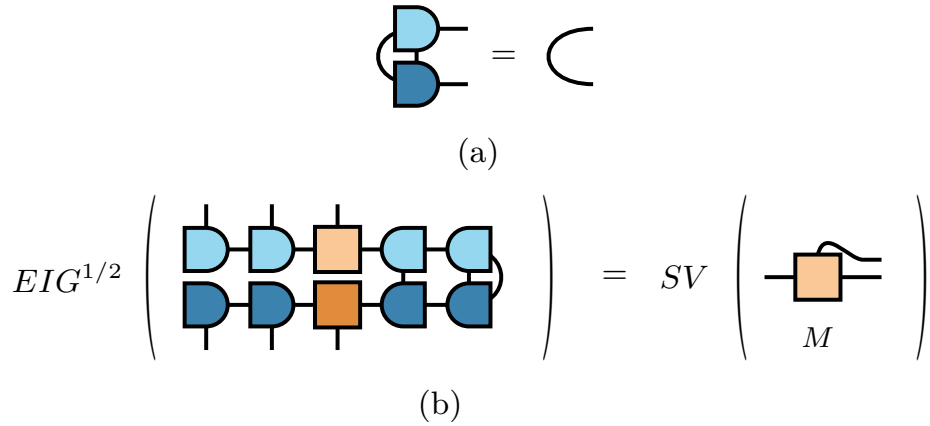


Figure 12: **Local RDM truncation in canonical form.** (a) Contracting a canonical tensor with its conjugate over the physical index and one virtual index gives the identity on the remaining index. The tensor and its conjugate are shown in lighter and darker shades. (b) Reduced density matrix at a cut, with the left block open and the right block traced. The isometries on either side of the orthogonality centre contract to identities, so the eigenvalues of the full **RDM** are the squared singular values of the centre tensor M .

This construction relies on two properties of the single-state problem. The truncated object is Hermitian and positive, so its eigenvalues are the squared Schmidt values and

their ordering is unambiguous. In addition, the Eckart–Young–Mirsky theorem provides a variational error bound. Neither property extends directly to a non-Hermitian transition matrix.

C.3 The reduced transition matrix and its truncation

In the transverse picture, the transfer matrix is non-Hermitian and its dominant left and right fixed points are distinct. The corresponding truncation object is the **RTM** in Eq. (11), which is constructed from both states. Its eigenvalues can be complex and do not admit the Schmidt interpretation above. Gauge transformations can also redistribute weight between components that contribute to $\langle L|R \rangle$ and those that do not [42]. Figure 13 compares the **RDM** and **RTM** at the same temporal cut.

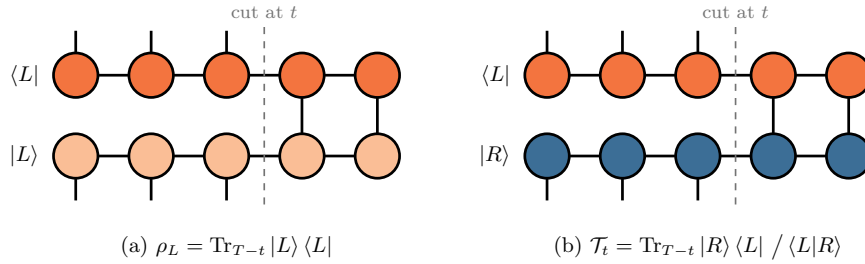


Figure 13: **Candidate truncation objects at a temporal cut.** Sites after the cut are traced over, while the physical legs before the cut remain open and form the matrix indices. (a) The **RDM** pairs a temporal state with its conjugate and is Hermitian and positive. (b) The **RTM** pairs the distinct fixed points $|R\rangle$ and $\langle L|$; it is generally non-Hermitian and can have complex eigenvalues.

The two objects transform differently under a gauge transformation X on the cut bond. The boundary **RDM** transforms by a congruence, $\rho_L \rightarrow X^\dagger \rho_L X$, which changes its spectrum for a non-unitary X . The **RTM** transforms by a similarity, $\mathcal{T}_t \rightarrow X^{-1} \mathcal{T}_t X$, and its eigenvalues are therefore invariant. Figure 14 shows these transformations at the network level. Consequently, the spectrum of a single temporal-state **RDM** depends on the representation, whereas the **RTM** spectrum does not.

The **RDM** scheme remains well defined in a fixed canonical gauge and can still be used as a numerical truncation method. The gauge argument instead identifies which spectrum is representation independent and can therefore be constrained by the **CFT**.

Both schemes must extract their truncation data from a local $\chi \times \chi$ environment matrix M_A , rather than diagonalising the exponentially large global object. For the Hermitian case, a bipartite cut gives

$$\rho_B = U_B M_A U_B^\dagger, \quad (34)$$

where U_B is unitary because it is constructed from left-orthogonal tensors. Equation (34) is a similarity transformation, so $\text{eig}(\rho_B) = \text{eig}(M_A)$. The local environment therefore contains the eigenvalues of the full **RDM**. For an **RTM** built from two different states, the same cut gives

$$\tau_B = U_B M_A V_B, \quad U_B \neq V_B^\dagger, \quad (35)$$

which is not a similarity transformation. Its eigenvalues are therefore not preserved: $\text{eig}(\tau_B) \neq \text{eig}(M_A)$. However, U_B and V_B remain isometries and preserve singular values,

$$\text{sv}(\tau_B) = \text{sv}(M_A). \quad (36)$$

Thus, the singular values of the local environment equal those of the full **RTM**. They are the invariant quantities accessible from the local reduction and provide the practical truncation criterion.

Singular values and eigenvalues nevertheless encode different information. They satisfy $\text{sv}(M) = \sqrt{\text{eig}(MM^\dagger)}$ and coincide for a Hermitian positive **RDM**, but not for a non-Hermitian **RTM**. The physical overlap is determined by the trace, $\text{Tr } \mathcal{T}_t = \sum_i \lambda_i$, which is a signed sum of eigenvalues and is not bounded directly by singular-value magnitudes. For example, eigenvalues $+1$ and -1 both have unit magnitude but cancel exactly in the trace.

The **RTM** truncation therefore lacks the variational bound available for an **RDM**. Its success here is empirical and relies on the structure of transfer matrices generated by time evolution; the cancellations that could invalidate the singular-value criterion are not observed in the calculations studied. We retain the **RDM** scheme as an independent numerical check.

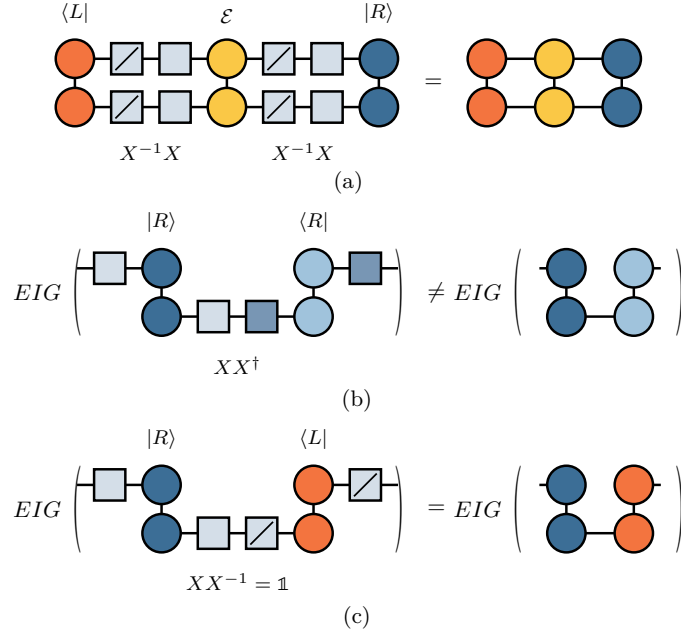


Figure 14: **Gauge transformation of the RDM and RTM.** (a) Inserting $X^{-1}X$ on a virtual bond changes the tensors but not the full contraction. In (b) and (c), the lower time site is contracted while the upper site remains open. (b) For the **RDM**, the gauge factors combine as $XX^\dagger \neq \mathbb{1}$ and produce the congruence $X\rho X^\dagger$, which changes the spectrum. (c) For the **RTM**, they combine as $XX^{-1} = \mathbb{1}$ and produce the similarity transformation $X\mathcal{T}_t X^{-1}$, which preserves the spectrum.

A generalised canonical form can be imposed on the pair $\langle L|, |R\rangle$ rather than on a state and its conjugate, as shown in Figure 15. Tensors away from the cut then contract to identities, and the singular values of the **RTM** follow from a local decomposition. The **RDM** and **RTM** routes consequently have comparable computational cost and differ primarily in the object they compress.

The transverse calculation depends on bilinear matrix elements rather than on the norm of either state separately. In particular, it involves quantities of the form $\langle L| \mathcal{O} |R\rangle$. A component of $|R\rangle$ contributes weakly if its overlap with $\langle L|$ is small, even when its weight in $|R\rangle$ is large, and the converse can also occur. The method therefore uses a biorthogonal decomposition of $|R\rangle \langle L|$ and retains the components with largest-magnitude

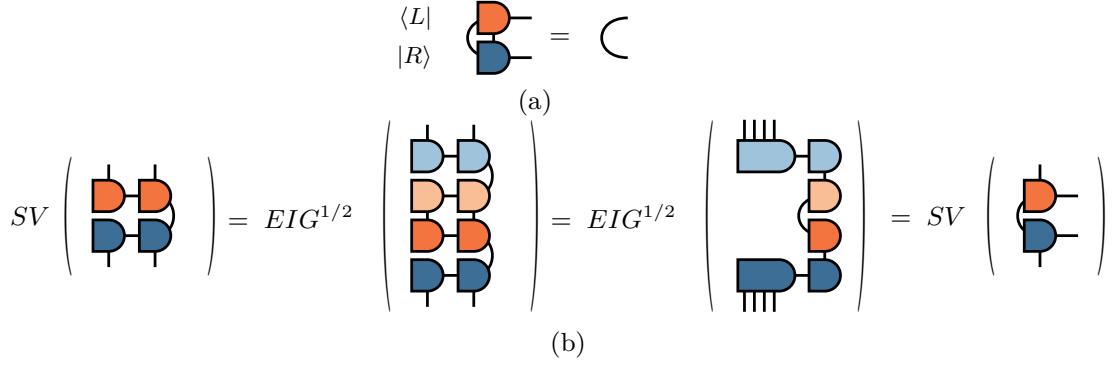


Figure 15: **Local RTM truncation in generalised canonical form.** (a) The contraction pairs a tensor of $\langle L|$ with the corresponding tensor of $|R\rangle$ rather than with its conjugate. (b) Imposing this condition along the chain reduces the environment to the tensors at the cut, whose singular values equal those of the RTM. The construction remains local but no longer has a variational error bound.

eigenvalues [27]. Since relevance is measured through the left–right overlap, this criterion can require fewer states than an RDM truncation at comparable accuracy.

The distinction between singular values and eigenvalues limits this argument. The contraction depends on the trace of $\mathcal{T}_t$, and hence on the signed sum of its eigenvalues, whereas the local reduction preserves its singular values. Large singular values do not preclude cancellation in the trace, as illustrated by eigenvalues $+1$ and -1 . Singular-value truncation of the RTM is therefore a physically motivated approximation rather than a rigorous bound.

A further approximation arises because each truncation uses the local environment available at that stage of the sweep rather than the uncontracted remainder of the network. Translation invariance reduces the resulting error because each subsequent bulk column applies the same operator and can regenerate components of the leading subspace that were previously discarded. Nevertheless, this approximation contributes to the greater numerical stability of the RDM route when the RTM result becomes noisy.

D The fermionic form of the NNN interactions

This appendix derives the fermionic representation of the model and shows that the NNN coupling generates genuine interactions for $p \neq 0$, removing the free-fermion integrability of the Ising chain.

The interaction structure becomes explicit after mapping the spin chain to fermions. In the convention of Eq. (13), with the transverse field along σ^x and the order parameter along σ^z , the JW transformation is [43, 44]

$$\sigma_j^x = 1 - 2n_j, \quad \sigma_j^z = (c_j + c_j^\dagger) K_j, \quad K_j = \prod_{k < j} (1 - 2n_k), \quad (37)$$

where c_j annihilates a spinless fermion, $n_j = c_j^\dagger c_j$, and the string K_j enforces the fermionic anticommutation relations.

The two terms present at $p = 0$ remain quadratic. The transverse field becomes a chemical-potential term, while the strings in the NN Ising coupling cancel and leave hopping and pairing terms. The resulting chain is a free-fermion model and can be diagonalised exactly [35]. At criticality, its continuum limit is the free Majorana CFT with $c = \frac{1}{2}$ [23].

The terms proportional to p generate quartic fermionic contributions. The self-dual $\sigma_j^x \sigma_{j+1}^x$ term becomes the density–density factor $(1 - 2n_j)(1 - 2n_{j+1})$, which is diagonal in the occupation basis. For the **NNN** bond, the Jordan–Wigner strings cancel only partially, leaving a parity factor on the intermediate site:

$$\sigma_i^z \sigma_{i+2}^z = (c_i^\dagger - c_i) (1 - 2n_{i+1}) (c_{i+2} + c_{i+2}^\dagger). \quad (38)$$

This term describes hopping and pairing between sites i and $i + 2$, with a sign determined by the occupation of the intermediate site. We refer to this process as parity-dressed hopping. The resulting quartic terms allow fermion scattering, so Wick’s theorem no longer reduces correlation functions to two-point functions [21]. The model is therefore interacting and does not admit a free-fermion solution for $p \neq 0$.

D.1 Jordan–Wigner and the four-fermion interaction

We now derive Eq. (38) from the transformation in Eq. (37). Pauli operators anticommute on the same site but commute on different sites, so local spin operators do not reproduce fermionic anticommutation at finite separation. The string K_j supplies the parity of all sites to the left of j and converts the spin algebra into the fermionic one. When the resulting Hamiltonian is quadratic, it can be diagonalised through a single-particle transformation into independent normal modes. This is the structure of the pure Ising chain and its free Majorana critical theory. The calculation below uses $(c + c^\dagger)(1 - 2n) = c^\dagger - c$ and $K_j^2 = 1$.

The transverse field becomes a chemical-potential term directly from Eq. (37). For the **NN** Ising bond, the two strings overlap on all but one site and cancel telescopically, leaving the quadratic hopping-and-pairing term

$$\sigma_j^z \sigma_{j+1}^z = (c_j^\dagger - c_j)(c_{j+1} + c_{j+1}^\dagger). \quad (39)$$

Together with the chemical-potential term, Eq. (39) gives the quadratic fermionic Hamiltonian of the pure Ising chain.

Both terms proportional to p are quartic. The self-dual $\sigma^x \sigma^x$ term becomes the density–density factor $(1 - 2n_j)(1 - 2n_{j+1})$. For the **NNN** bond, the strings cancel on the common sites, while the intermediate site contributes one uncanceled factor $(1 - 2n_{i+1}) = (-1)^{n_{i+1}}$. This gives Eq. (38), in which the sign of hopping and pairing between i and $i + 2$ depends on the intermediate occupation.

E Extraction and equilibrium details

This appendix gives the equilibrium measurements and fitting conventions used for the numerical values quoted in the main text. These include the finite-size test of the universality class, the sound velocity, and the branch constant in the eigenvalue fit.

The leading boundary exponent distinguishes the two boundary conditions of Section 2.2. At the integrable point, fitting Eq. (7) to the first spectral gap over $T = 3\text{--}9$ gives $x_1 = 0.499$ for the free boundary state $|X+\rangle$ and $x_1 = 1.988$ for the fixed state $|Up\rangle$, in agreement with the exact values $\frac{1}{2}$ and 2. Figure 16 shows both.

E.1 Finite-size scaling of the mid-chain entropy

Section 4.1 extracts the central charge from the entanglement profile at a fixed system size. An independent estimate follows from the size dependence of the entropy at the

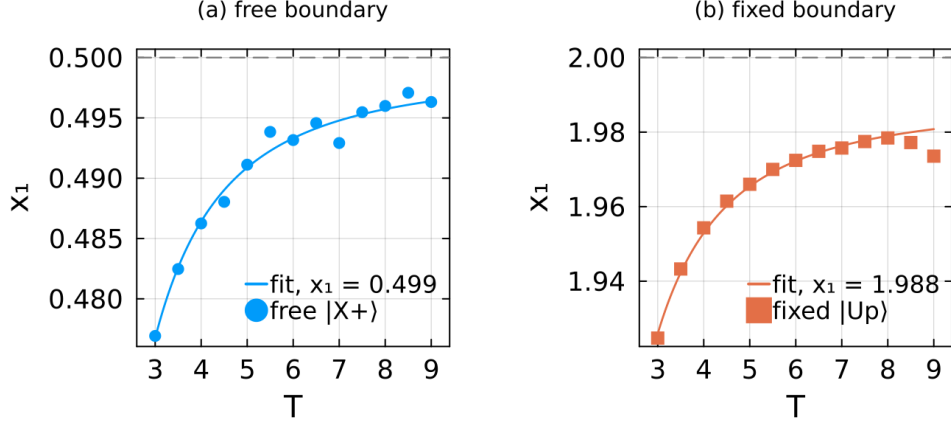


Figure 16: **The boundary exponent at the integrable point.** The first spectral gap converted into a dimension, $x_1 = v T \text{Im}(\lambda_1 - \lambda_0)/\pi$, against T for the free and fixed boundary states. Panel (a) is the free boundary state $|X+\rangle$ and panel (b) the fixed state $|Up\rangle$, side by side, each on its own scale. Dashed lines mark the exact values $\frac{1}{2}$ and 2. Both approach their target from below as T grows, and the fits of Eq. (7) over the same thirteen evolution times give 0.499 and 1.988. The window here is shorter than the one behind Table 1, since it is cut to the range the fixed ladder covers so that the two boundary conditions are fitted over identical times; this is why the free value differs slightly from the one quoted there.

mid-chain cut. At this cut, the chord variable is maximal, $W(L/2, L) = \ln(2L/\pi)$, and Eq. (14) reduces to

$$S_{\text{vN}}(L/2) = \frac{c}{6} \ln L + k'. \quad (40)$$

A linear fit of the mid-chain entropy against $\ln L$ therefore has slope $c/6$. For $N = 40$ –200, this procedure gives $c \approx 0.51$ at both $p = 0.1$ and $p = 0.5$, consistent with the profile fit in the main text and with the Ising value. These calculations use fifteen DMRG sweeps, a bond-dimension ramp up to 400, and a truncation cutoff of 10^{-10} . The $N = 400$ chord fit in Section 4.1 uses fifty sweeps, a maximum bond dimension of 1000, and a cutoff of 10^{-12} . Both calculations begin with noise-assisted sweeps.

The same ground states provide a comparison with the Rényi-2 entropy used in the temporal calculations. Its mid-chain coefficient is $c/8$, but a linear fit gives $c \approx 0.58$, above the Ising value. Rényi entropies with $n > 1$ contain oscillatory finite-size corrections that decay algebraically with subsystem size [34]; omitting them biases the fitted slope upwards. The corresponding temporal correction scales as $\Delta S_n \propto T^{-1/n}$ [19] and is included explicitly in Section 7.1. We do not fit the equilibrium corrections because the von Neumann entropy already provides a stable estimate when the full Schmidt spectrum is available. Figure 17 compares the two finite-size results.

E.2 The sound velocity from the equilibrium dispersion

We now give the construction and validation of the sound-velocity estimates used in Section 4.2.

We use finite periodic chains, for which momentum is a good quantum number, and diagonalise the Hamiltonian in Eq. (13) exactly at $\lambda = 1$. Each eigenstate is classified by its momentum, obtained from the one-site translation operator, and by its spin-flip parity. Since numerical diagonalisation returns arbitrary mixtures within degenerate subspaces, momentum and parity are resolved jointly before the states are labelled. In the magnetic,

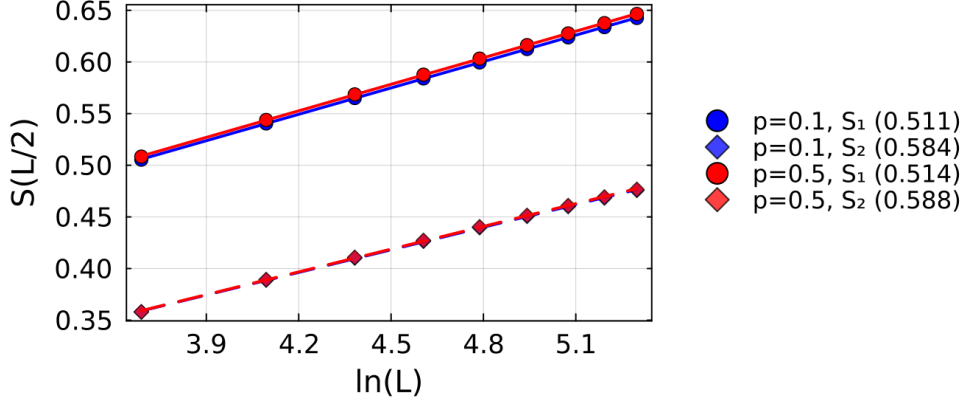


Figure 17: **Finite-size scaling of the mid-chain entropy.** Results for $p = 0.1$ and $p = 0.5$. The von Neumann entropy (solid lines), with slope $c/6$, gives $c \approx 0.51$. The Rényi-2 entropy (dashed lines), computed from the same ground states and fitted with slope $c/8$, gives $c \approx 0.58$ because of its larger finite-size corrections.

odd-parity sector, we identify the lowest states at momenta 0 and k and compute the velocity from their dispersion:

$$v = \frac{N}{2\pi} \frac{E_1(k) - E_1(0)}{|k|}, \quad (41)$$

where k is the smallest non-zero momentum on the ring. Equation (41) is the finite-size form of the low-momentum dispersion $\omega = v|k|$. The allowed momenta are separated by $2\pi/N$, so the prefactor converts the energy difference between adjacent momentum sectors into the dispersion slope.

Four checks validate this construction. At $p = 0$, the estimate approaches the exact value $v = 2$ and reaches 1.990 at $N = 18$. The independently constructed ring Hamiltonian agrees with the operator used elsewhere in the code to within 7×10^{-14} over the full spectrum at $N = 8$. Sector-resolved and sector-independent calculations give the same velocity at every p . Finally, inserting the measured $v(p)$ into the Casimir formula of Ref. [20] gives $c = 0.504$ at $p = 0$ and $c = 0.525$ at $p = 0.5$ for $N = 16$, consistent with the published ground-state results.

The accessible ring sizes remain far from the thermodynamic limit. At $p = 0$, the estimate is 1.9899 at $N = 18$, compared with the exact value 2. The finite-size error follows a regular $1/N^2$ dependence: multiplying it by N^2 gives values between 3.27 and 3.28 for $N = 10$ –18. Directly reaching three-digit accuracy would require approximately $N = 57$, which is beyond exact diagonalisation, so we extrapolate this leading correction.

At each coupling, we fit $v(N) = v_\infty + C/N^2$ and use v_∞ in the conformal analysis. Table 2 lists the data and fit parameters, and Figure 18 shows the extrapolations. The residuals remain below 2.8×10^{-2} for velocities between approximately two and eleven. At $p = 0$, the extrapolation gives 1.99994, within 6×10^{-5} of the exact value, although the $N = 18$ result differs by 1.0×10^{-2} . The finite-size coefficient grows in magnitude from $C = -3.27$ at $p = 0$ to $C = -140.7$ at $p = 1.5$, making the extrapolation increasingly important with frustration. Using these coupling-dependent velocities, rather than fixing $v = 2$, is necessary to recover the expected boundary dimensions.

Over the range studied, the extrapolated velocity is approximately linear in p : the fit $v_\infty = K_0 + K_1 p$ gives $v_\infty = 2.02 + 6.42p$, with residuals below 0.023. We use this

p	$N = 10$	$N = 12$	$N = 14$	$N = 16$	$N = 18$	v_∞	C
0.0	1.9673	1.9772	1.9833	1.9872	1.9899	1.99994	-3.27
0.1	2.5707	2.6007	2.6191	2.6312	2.6396	2.67016	-9.97
0.2	3.1565	3.2073	3.2387	3.2594	3.2738	3.32561	-16.96
0.3	3.7247	3.7972	3.8423	3.8722	3.8931	3.96723	-24.34
0.4	4.2756	4.3707	4.4305	4.4703	4.4981	4.59585	-32.16
0.5	4.8096	4.9284	5.0038	5.0543	5.0897	5.21214	-40.46
1.0	7.2534	7.5020	7.6674	7.7824	7.8653	8.12650	-88.23
1.5	9.4140	9.7946	10.0594	10.2504	10.3921	10.79924	-140.69

Table 2: **Finite-size velocities and thermodynamic extrapolations.** Values of $v(p)$ at $\lambda = 1$ for periodic chains of $N = 10$ –18, together with the parameters of the fit $v(N) = v_\infty + C/N^2$. The largest residual increases from 1.1×10^{-5} at $p = 0$ to 2.8×10^{-2} at $p = 1.5$. The conformal analysis uses the extrapolated values v_∞ .

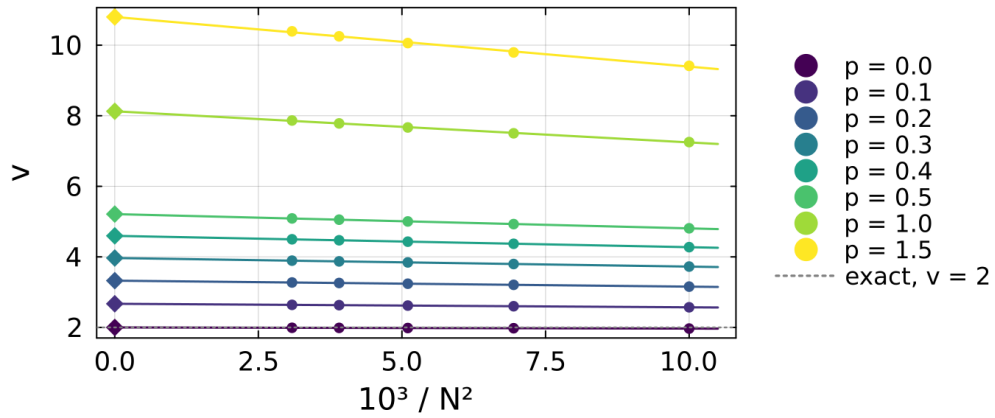


Figure 18: **Extrapolation of the sound velocity.** Finite-size estimates of v as a function of $1/N^2$, with fitted lines and thermodynamic intercepts (diamonds). At $p = 0$, the extrapolated value agrees with the exact result $v = 2$, shown by the dotted line.

relation only as a description of the data. Fits at fixed system size retain a noticeable N dependence, with (K_0, K_1) changing from $(2.00, 5.68)$ at $N = 10$ to $(2.01, 6.13)$ at $N = 16$. Moreover, the linear-fit intercept is less accurate at $p = 0$ than the direct thermodynamic extrapolation. All conversions in this thesis therefore use the separately extrapolated $v_\infty(p)$ rather than the linear fit.

We also verify the velocity dependence using the anisotropic XY chain, for which $v(\gamma) = 2\gamma$ is exact. At $\gamma = 0.5, 1, 1.5$, the measured first-gap slopes are $a_1 = 1.481, 0.782, 0.520$, compared with the exact values $\pi/(4\gamma) = 1.571, 0.785, 0.524$. Using the corresponding velocities gives $x_1 = 0.471, 0.498, 0.496$. Fixing $v = 2$ would instead impose $a_1 = 0.785$ for all three anisotropies and would disagree with two of the measured slopes. The extraction recovers $x_1 = \frac{1}{2}$ only when the correct velocity is supplied.

E.3 The entropy profiles at stronger coupling

Section 7.1 shows the entropy profiles at $p = 0.1$. Figure 19 gives the same comparison at the remaining couplings, each to the longest evolution time reached. The real part follows the chord curve at $c = \frac{1}{2}$ throughout, and the imaginary part is constant across the interior of the strip and lies above $\pi c/16$ by the finite-time correction, exactly as at the two couplings shown in the main text. The number of evolution times available falls as the coupling grows, since the cost of each one rises with p ; at $p = 0.5$ only four are available, which is too few to constrain the extrapolation.

E.4 Fit windows and the controls behind the quoted numbers

The central charges in Table 1 are obtained from least-squares fits. We specify here the fitting windows, the fixed parameters, and the numerical controls used to assess their stability.

The start of the fitting window is not unique. Moving it to later times reduces the number of data points and increases the fitted central charge because the $1/T$ and $1/T^2$ contributions in Eq. (6) become difficult to distinguish over a short interval. Varying the starting time changes c by 0.014 at $p = 0$, 0.024 at $p = 0.1$, and 0.21 at $p = 0.3$. We report the widest window that satisfies the consistency criterion, since it contains the most data and best separates the correction terms. The upper end is set by the loss of a constant phase increment in μ_0 . At $p = 0.5$, for example, the increment holds to a few parts in 10^4 across the fitted window and becomes irregular beyond it. This criterion determines the windows in Table 1.

We fix the branch constant in Eq. (6) rather than fitting it. Its value follows from $\lambda_0 = \log(-\mu_0)$, and unconstrained fits over sufficiently long windows give $B/\pi = -1.000$ at every coupling. Fixing it removes a parameter that short windows cannot determine. The phase is unwrapped over the complete time series before selecting a fitting window; otherwise the absolute branch information would be lost.

We use three numerical controls. Increasing the bond dimension to $\chi = 128$ changes the imaginary plateau by approximately 2% and $|\mu_0|$ by less than 10^{-4} , but shifts the real-part chord slope by up to 0.12. This motivates extracting the entropy result from the plateau. Halving the Trotter step while doubling N_β to keep β_0 fixed changes the plateau by less than 1% within the reliable window and by 18% near its boundary. Repeating one time point with five independent random seeds produces a 3% spread in the plateau and a 12% spread in the chord slope, whereas $|\mu_0|$ agrees within 1.3×10^{-4} . The spectral estimates are therefore substantially less sensitive to the random seed. Finally, the sound velocity

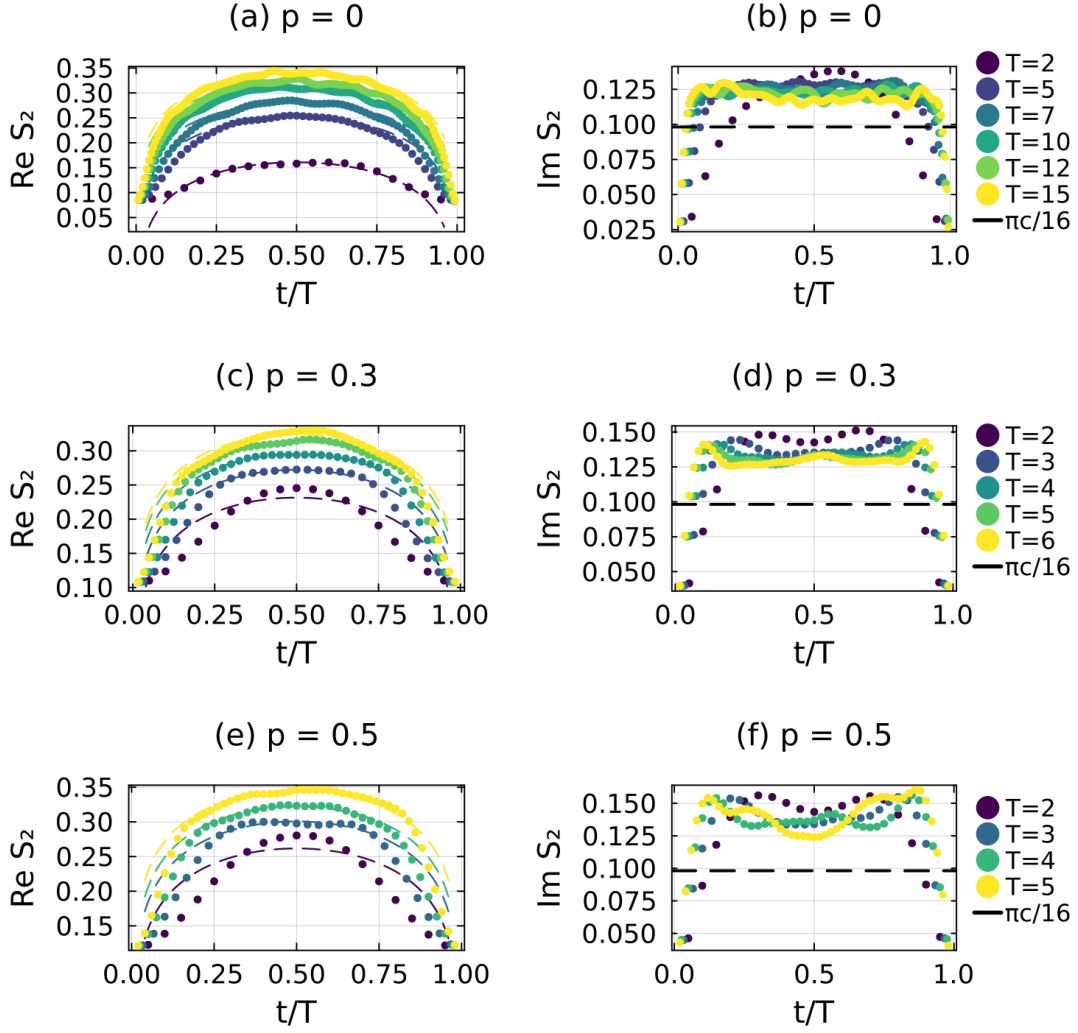


Figure 19: **The generalized temporal entropy at the remaining couplings.** The comparison of Figure 8 at $p = 0, 0.3$ and 0.5 . Each row shows up to six evolution times, evenly spaced across the range the calculation reached. (a, c, e) $\text{Re } S_2$ against t/T , with the dashed curves showing Eq. (5) at $c = \frac{1}{2}$ and the non-universal constant matched to each profile. (b, d, f) $\text{Im } S_2$, with the dashed line at $\pi c/16 = 0.098$ for $c = \frac{1}{2}$. The separation between the data and that line is the finite-time correction, as in Figure 8. The $p = 1.0$ profiles use the halved Trotter step of Section 7.

enters linearly, and its extrapolation gives 1.99994 at the integrable point, compared with the exact value 2.

The entropy-route central charges of Section 7.1 also depend on the finite-time correction model. We compare a constant, $a + bT^{-1/2}$, $a + b/T$, and $a + b/T + c/T^2$ using the corrected Akaike information criterion. Over $T = 2\text{--}24$, the constant model is disfavoured by a difference of 21. The $T^{-1/2}$ and two-term models differ by less than 1 and give $c = 0.506$ and 0.511 , respectively. Over the shorter interval $T \leq 11$, the data do not distinguish the four models, demonstrating the need for the extended time series.

F Construction of the time-evolution MPO

This appendix describes the time-evolution MPO used in Section 5. We give the practical exponentiation procedure, outline the two available constructions, record the explicit second-order tensor, and compare their numerical accuracy.

F.1 How the exponentiation is carried out

We perform the exponentiation with `ITensorExpMPO.jl` [45], which implements the construction of Ref. [41] using `ITensors.jl` [46]. The Hamiltonian is supplied as a symbolic sum of operator strings, so the FSM in Section 5.1 need not be constructed manually. At each bond, the package collects the terms crossing that bond and compresses their coefficient matrix with an SVD. The retained components form the minimal set of memory channels needed to represent the interactions. For Eq. (13), these are the three channels in Eq. (15). The resulting blocks A , B , C , and D are then passed to the routine that constructs the local evolution tensor.

The original package provides the first- and second-order operators of Ref. [41]. We added an implementation of the second-order construction of Ref. [40], including the required enlargement of the virtual bond from $1 + \chi$ to $1 + \chi + \chi^2$. This implementation is available in a fork of the package [47]; the repositories contain the software documentation.

F.2 The two constructions in outline

Both constructions address the symmetric cross terms generated at second order. Directly expanding W_H^2 includes every ordered pair of Hamiltonian blocks, so terms such as CD and DC must both be retained to preserve the Hermiticity inherited from H . In the construction of Ref. [41], the blocks are encoded in an auxiliary space of two hard-core bosons. The auxiliary exponential terminates once both modes are occupied, and projection onto the four bosonic states gives the blocks of the evolution MPO. Retaining only disjoint-support terms gives the first-order operator W^I . Adding single-site overlaps gives W^{II} , which is second order for Hamiltonians containing only NN interactions. Both use virtual bond dimension $1 + \chi$.

The construction of Ref. [40] instead organises the expansion into connected clusters of overlapping terms. Disjoint terms commute and factorise, so only connected clusters require an explicit approximation. Retaining clusters up to size n gives a one-step error of order δt^{n+1} . The second-order operator used here introduces a doubled memory sector, increasing the virtual bond dimension to $1 + \chi + \chi^2$. This sector represents the three-site connected clusters generated by the NNN interaction. The resulting local tensor is given below.

In each scheme, the complex step $\tau = -i\delta t$ is distributed asymmetrically between the interaction blocks. The initiator C is multiplied by $t_C = \sqrt{\delta t}$ and the terminator B by

$t_B = \tau/t_C$. Their product retains the required factor τ , while the individual memory-channel amplitudes remain numerically balanced.

F.3 The second-order $VD2$ blocks

We now give the explicit local tensor of the second-order **MPO**, denoted $VD2$, used in the simulations. The expressions follow Ref. [40] and are specialised to the **FSM** blocks of the present Hamiltonian.

We start from the four blocks A, B, C, D of the Hamiltonian MPO W_H (Eq. (15)): D is the on-site term, C initiates an interaction, B terminates one, and A propagates it. Following the time-step split above, we scale them as

$$\tilde{D} = \tau D, \quad \tilde{C} = t_C C, \quad \tilde{B} = t_B B, \quad \tilde{A} = A, \quad t_C = \sqrt{\delta t}, \quad t_B = \tau/t_C, \quad \tau = -i\delta t. \quad (42)$$

The expansion of $e^{\tau W_H}$ is expressed in terms of symmetrised products of overlapping blocks. We denote by $\{X_1 \dots X_m\}$ the sum over all distinct orderings of the operator-valued blocks. For example, $\{XY\} = X \otimes Y + Y \otimes X$ and $\{XXY\} = XXY + XYX + YXX$. Here $\otimes$ denotes the Kronecker product on the virtual memory index and ordinary operator multiplication on the physical index. The explicit prefactors below include the factorial weights from the exponential series.

The local tensor has a 3×3 block structure. Indices 1, 2, and 3 label the scalar, single-memory, and doubled-memory sectors, respectively, giving the virtual bond dimension $1 + \chi + \chi^2$:

$$W^{VD2} = \begin{pmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{pmatrix}, \quad (43)$$

with

$$W_{11} = \mathbb{1} + \tilde{D} + \frac{1}{2}\{\tilde{D}\tilde{D}\} + \frac{1}{6}\{\tilde{D}\tilde{D}\tilde{D}\}, \quad (44)$$

$$W_{12} = \tilde{C} + \frac{1}{2}\{\tilde{C}\tilde{D}\} + \frac{1}{6}\{\tilde{C}\tilde{D}\tilde{D}\}, \quad W_{13} = \{\tilde{C}\tilde{C}\} + \frac{1}{3}\{\tilde{C}\tilde{C}\tilde{D}\}, \quad (45)$$

$$W_{21} = \tilde{B} + \frac{1}{2}\{\tilde{B}\tilde{D}\} + \frac{1}{6}\{\tilde{B}\tilde{D}\tilde{D}\}, \quad W_{31} = \frac{1}{2}\{\tilde{B}\tilde{B}\} + \frac{1}{6}\{\tilde{B}\tilde{B}\tilde{D}\}, \quad (46)$$

$$W_{22} = \tilde{A} + \frac{1}{2}(\{\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{D}\}) + \frac{1}{6}(\{\tilde{C}\tilde{B}\tilde{D}\} + \{\tilde{A}\tilde{D}\tilde{D}\}), \quad (47)$$

$$W_{23} = \{\tilde{A}\tilde{C}\} + \frac{1}{3}(\{\tilde{A}\tilde{C}\tilde{D}\} + \{\tilde{C}\tilde{C}\tilde{B}\}), \quad W_{32} = \frac{1}{2}\{\tilde{A}\tilde{B}\} + \frac{1}{6}(\{\tilde{A}\tilde{B}\tilde{D}\} + \{\tilde{B}\tilde{B}\tilde{C}\}), \quad (48)$$

$$W_{33} = \{\tilde{A}\tilde{A}\} + \frac{1}{3}(\{\tilde{A}\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{A}\tilde{D}\}). \quad (49)$$

The block W_{11} contains the on-site evolution through third order in the Taylor expansion. The blocks W_{12} and W_{13} describe outgoing single- and doubled-memory sectors, while W_{21} and W_{31} describe the corresponding incoming sectors. The lower-right blocks combine propagation through $\tilde{A}$ with the required symmetrised products. In particular, the doubled-memory sector represents connected clusters spanning three sites. These are the **NNN** overlaps omitted by W^{II} , and their inclusion makes $VD2$ second order for the present Hamiltonian.

F.4 Comparing the two constructions in practice

We compare the two constructions through their violation of norm conservation. Exact real-time evolution preserves $\|\psi\|$, so the norm drift produced by successive approximate layers provides a direct measure of the propagation error. We apply each operator to $|X+\rangle^{\otimes N}$ and monitor the norm after each step.

Applying eight layers at $\delta t = 0.05$ to a chain of $N = 100$ sites at $p = 0.1$ gives the following norms:

	W^I	W^{II}	$VD2$
virtual bond	4	4	13
$\ \psi\ $ after 1 layer	1.3514	1.1341	0.99995
$\ \psi\ $ after 8 layers	12.6146	3.0181	0.99923

The first-order operator W^I increases the norm by more than an order of magnitude after eight layers. The operator W^{II} , which is second order for a strictly **NN** Hamiltonian, increases it by a factor of three because it omits connected clusters generated by the **NNN** interaction. Over the same interval, $VD2$ keeps the norm within 8×10^{-4} of unity.

The larger virtual bond of $VD2$ is therefore required to control the time-step error. Reducing δt sufficiently to compensate for a first-order propagator would increase the temporal chain length and introduce additional truncations. A second-order operator at moderate δt provides the more efficient representation for the transverse calculation.

G Implementation of the block power method

This appendix describes the implementation of the block power method introduced in Section 6.1. The core routine, `block_transfer_eigs`, is built on `ITransverse.jl` [28]. The code, analysis notebooks, and cluster drivers are available in the thesis repository [48].

G.1 The regularised solve and the left pairing

Near a degeneracy, the columns of the right block become nearly linearly dependent and the metric S in Eq. (19) becomes ill conditioned. Direct solution of the generalised eigenvalue problem $Mv = \theta Sv$ can then produce divergent or unstable values. We instead use the pseudoinverse of S :

$$W = S^+ M, \quad W = V \Theta V^{-1}, \quad (50)$$

The pseudoinverse uses a relative tolerance of 10^{-12} , so directions that have become numerically dependent are projected out rather than inverted. After sorting by modulus, the diagonal entries of Θ estimate $\mu_0, \dots, \mu_{k-1}$, and the columns of V give the right mixing coefficients.

The left block requires coefficients paired with the same eigenvalues. A separate diagonalisation of the transposed pencil is unreliable within a quasi-degenerate cluster because its eigenvalues can be returned in a different order, making the left-right pairing ambiguous. The coefficients can instead be obtained directly from the right decomposition. Defining $y^T = u^T S$ transforms the left problem into $y^T W = \theta y^T$, so the vectors y are the rows of V^{-1} and

$$u_j = (S^+)^T (V^{-1})^T e_j, \quad (51)$$

is paired with θ_j by construction. The resulting coefficients are biorthogonal in the metric S , satisfying $u_i^T S v_j = \delta_{ij}$.

The **MPS** linear combinations used for de-mixing are formed by exact direct-sum addition and compressed after the sum is complete. The default density-matrix addition becomes unstable when the vectors are nearly parallel, as occurs within a tightly clustered spectrum.

G.2 The eigenvalue-only mode

The eigenvalue-only mode replaces the de-mixing coefficients by orthonormal bases. After sorting the Ritz values by modulus, we apply QR decompositions to the coefficient matrices:

$$V \rightarrow Q_V, \quad U \rightarrow Q_U, \quad Q^\dagger Q = \mathbb{1}, \quad (52)$$

The blocks are then rotated with Q_V and Q_U rather than V and U . The first j columns of each orthonormal matrix span the same ordered subspaces as the corresponding coefficient matrix, giving an ordered-Schur-like basis. This basis change does not alter the Ritz values. Under invertible recombinations B and C , the projected matrices transform as $S \rightarrow B^\top S C$ and $M \rightarrow B^\top M C$, leaving the solutions of the pencil unchanged. It does, however, improve the conditioning of the subsequent truncation: $\text{cond}(V)$ increases as the internal spectral gap closes, whereas Q_V and Q_U remain unitary. The two modes can therefore differ only through truncation. At $p = 0.1$ and $\chi = 64$, independent time series in the two bases agree in the relative moduli of the two monitored eigenvalues to 10^{-7} – 10^{-4} over $T = 2$ – 8 . The unmonitored eigenvalues agree to 10^{-3} – 10^{-2} .

Three safeguards are applied in both modes. Each block member is normalised after de-mixing, and any vector whose norm underflows or overflows is replaced by a new random vector. We also monitor the condition number of S . When two block directions become numerically dependent, the final member is replaced and biorthogonalised against the remaining vectors using the bilinear form. Convergence is assessed from the leading Ritz values, with the first two monitored in production. Values are matched to their nearest predecessors before computing their change, preventing swaps in modulus order within a near-degenerate cluster from appearing as discontinuities. The iteration terminates when the matched change falls below 10^{-6} or ceases to improve for a prescribed number of iterations.

G.3 Validation against exact diagonalisation

At short evolution times, the rotated transfer operator has dimension below a few thousand and can be contracted into a dense matrix for exact diagonalisation. This comparison tests the operator construction, rotation, block iteration, and projected pencil against a result without **TN** truncation. Table 3 shows the results for $k = 4$. The two leading Ritz values agree with exact diagonalisation to between 10^{-8} and 10^{-13} .

operator	dimension	exact $ \mu_0 , \mu_1 $	error
ANNNI-type, $VD2$, $p = 0.1$	343	0.9598, 0.3263	1.4×10^{-8}
ANNNI-type, W^{II} , $p = 0.1$	27	0.9758, 0.3292	4.6×10^{-13}
critical Ising	256	1.5488, 0.0611	4.2×10^{-10}

Table 3: **Validation against exact diagonalisation.** Comparison between the block method and dense diagonalisation of the same rotated transfer operator at short evolution times. The reported error is the largest absolute deviation in the moduli of the two leading Ritz values.

G.4 Computational cost

Each block-power iteration contains application, projection, and de-mixing steps. The application step performs $2k$ **tMPO**–**tMPS** products, increasing the temporal bond dimension from χ to $D\chi$ before compression. The projection step evaluates the k^2 bilinear overlaps defining S and M ; it does not increase the state dimensions, and the remaining

$k \times k$ dense algebra is negligible. The de-mixing step forms $2k$ linear combinations of k states. Exact direct-sum addition temporarily increases the bond dimension to $k\chi$ before compression, so this step determines both the peak memory and most of the runtime. For a representative production calculation with $\chi = 64$, $k = 4$, and $p = 0.1$, de-mixing and compression account for approximately 85% of each iteration.

We reduce this cost in two ways. First, partial direct sums are compressed progressively back to the production bond dimension, while the final addition remains exact. Second, the allowed stagnation period is shortened because late-time calculations can remain close to the convergence threshold for many iterations. Together, these changes reduce the runtime of the longest time points by approximately a factor of two and change $|\mu_0|$ by only 4×10^{-6} relative.

Each truncation uses the current states without information from the network columns that remain to be contracted. A discarded component could therefore become relevant at a later step. In the translation-invariant bulk, however, each subsequent column applies the same operator $\mathcal{E}$ and can regenerate components belonging to the leading invariant subspace.

G.5 Production options

A model-independent builder constructs the one-step propagator as an **MPO** (Appendix F), performs the 90° rotation from the spatial virtual bond to the temporal physical index, and returns the transfer-matrix **MPO** together with a seed **tMPS** carrying the required index structure. Before iteration, the seed tensors are replaced by random complex tensors because a symmetry-restricted seed can converge to a subdominant sector. The temporal physical dimension is inferred from the operator, allowing the same implementation to treat W^I , W^{II} , $VD2$, and different models.

The following options modify the numerical procedure without changing its target fixed point.

Warm starting. The converged block at one evolution time is re-indexed onto the longer temporal lattice of the next point and extended with a small random tail. The iteration then needs primarily to relax the new sites. The bond cap increases in steps of two up to χ , while the truncation cutoff remains at 10^{-8} for the first forty iterations and then decreases to 10^{-10} . Initialising consecutive times from the same leading subspace also improves continuity in the eigenvalue selection.

Block-size escalation. At stronger next-nearest-neighbour coupling, a $k = 4$ block can be occupied entirely by one near-degenerate cluster, leaving the subleading eigenvalue required by Eq. (7) outside the computed subspace. The driver then increases the block size as $k = 4 \rightarrow 6 \rightarrow 8$, initialising each larger calculation from the converged smaller block. The additional cost is therefore incurred only when required by the spectrum.

Truncation scheme. The calculation can use either **RTM** or **RDM** truncation (Section 3.4). The **RTM** is the default, while the better-conditioned **RDM** provides a fallback and control.

Checkpointing. Each converged (p, T) point is stored immediately in a parameter-indexed cache. An interrupted calculation therefore loses at most the current point. A separate checkpoint stores the latest converged block and permits warm restarting after resubmission.

G.6 Selecting the dominant eigenvalue

As stated in Section 6.2, we select the eigenvalue of largest modulus and use continuity with the preceding time point to resolve differences below one per cent. If the selected $|\mu_0|$ changes by more than a prescribed fraction relative to the previous accepted value, the point is flagged as potentially unconverged. The stored spectrum is then reanalysed by restricting the candidates to a five-per-cent window around the previous modulus and selecting the dominant eigenvalue within that interval. The reference is updated sequentially from the last accepted point. For the case with an independently converged comparison, this correction agrees with the repeated calculation to 3×10^{-5} relative. Subleading eigenvalues are ordered by their phase distance from μ_0 and matched continuously to the conformal tower.

When a numerical limitation appears, we repeat representative calculations at half the bond dimension, with a tighter truncation cutoff, and with RDM instead of RTM truncation. We interpret a feature as intrinsic to the transfer matrix only when it persists under these changes.

G.7 Why eigenvalues outlive eigenvectors

The loss of reliability cannot be attributed to a fixed threshold in the phase rigidity. The rigidity of the dominant eigenvector pair decreases geometrically with T and does so more rapidly as the next-nearest-neighbour coupling grows: at $T = 14$, it is 1.3×10^{-4} for $p = 0$ and 1.6×10^{-10} for $p = 0.1$. However, its value at the last usable time spans four orders of magnitude across the couplings studied. Moreover, the numerical failures are intermittent even though the rigidity decreases monotonically. For example, at $p = 0$, the calculations at $T = 17$ and $T = 22$ are unreliable, whereas those at $T = 18$ – 21 and $T = 23$ remain usable. Low rigidity therefore correlates with reduced numerical stability but does not by itself determine whether a calculation fails.

The different numerical stability of eigenvalues and eigenvectors follows from first-order perturbation theory for non-Hermitian matrices [49]. We perturb $\mathcal{E}$ by $\delta\mathcal{E}$ and expand the perturbed right eigenvector as $|R_i\rangle + \sum_{j \neq i} a_{ij} |R_j\rangle$. Projection onto $\langle L_i|$ gives the eigenvalue shift,

$$\delta\mu_i = \frac{\langle L_i | \delta\mathcal{E} | R_i \rangle}{\langle L_i | R_i \rangle}, \quad (53)$$

while projection onto $\langle L_j|$ for $j \neq i$ gives the eigenvector-mixing coefficients,

$$a_{ij} = \frac{\langle L_j | \delta\mathcal{E} | R_i \rangle}{(\mu_i - \mu_j) \langle L_j | R_j \rangle}. \quad (54)$$

Equation (53) contains no spectral gap, so the eigenvalue remains stable while the overlap $\langle L_i | R_i \rangle$ remains finite. By contrast, Eq. (54) contains the denominator $\mu_i - \mu_j$. As two eigenvalues approach, their eigenvectors become increasingly sensitive to perturbations and the de-mixing of individual block members becomes ill conditioned.

The overlap appearing in Eq. (53) is quantified by the phase rigidity of a biorthogonal pair,

$$r_j = \frac{|\langle L_j | R_j \rangle|}{\|L_j\| \|R_j\|}, \quad (55)$$

which is the normalised overlap of the left and right eigenvectors associated with the same eigenvalue. It equals one for a Hermitian matrix and approaches zero as their biorthogonal overlap decreases. A small rigidity therefore increases the sensitivity of an eigenvalue

associated with an individual eigenvector. The leading invariant subspace can nevertheless remain well conditioned when it is separated from the rest of the spectrum. The eigenvalue-only mode exploits this distinction: it propagates an orthonormal basis of the leading subspace and obtains the Ritz values from the projected pencil without assigning each basis vector to a particular eigenvector. Its improved stability arises from avoiding the ill-conditioned de-mixing required by the eigenvector calculation.

The ratio $|\mu_1/\mu_0|$ controls the suppression of the first subleading contribution in Eq. (9), but it does not determine convergence of the block iteration. Convergence instead depends on the separation between the complete leading subspace and the remainder of the spectrum. Near a degeneracy within the retained block, $|\mu_1/\mu_0|$ can approach one while the subspace remains well separated and its Ritz values continue to converge.